DANFYSIK

System 8800

User's Manual

MPS 883

OUTPUT DATA : XXXA / XXV

MPS 883

Project no XXXXXX

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1 Introduction and specifications

1.1 General introduction

The System 8800 Precision Magnet Power Supplies are DC constant current output Power Supplies designed for applications requiring very high stability and low noise combined with reliability and ease of operation.

The System 8800 is aimed at precision spectroscopy and ion beam transport applications. It is the result of an intensive development effort at Danfysik based on more than 30 years of experience in delivering precision DC Power Supplies to industrial and research laboratory users around the world.

The System 8800 is available as a range of power, control and interface modules which can be configured to meet specific application requirements with guaranteed performance.

* Current stability of 25 ppm.

* Power outputs from 5 kW to 25 kW per unit. 4 units can be mounted in one 19" rack, and many racks may be coupled in parallel for higher currents.

* Output current up to maximum 600 A.

* Precision current transducer ULTRASTAB (DCCT = DC Current Transformer) to achieve new performance levels for stability and linearity over very wide current ranges.

* Optically isolated digital control to eliminate ground loops and conducted noise. The control and interface electronics modules are isolated from the power modules by an electrostatic and thermal shielding wall.

1.2 Specifications

DC OUTPUT RATING

Power range ..... XXkW
Current range ..... XXA
Max.Voltage. 20Vdc

CURRENT STABILITY CLASS

Current control range..... 0.2-100%
Nominal operating range 5-100%

8 hours stability in nominal operating range
(ppm of max current)..... ±25 ppm

30 min. stability in nominal operating range
(ppm of max current)..... ± 10 ppm

PERFORMANCE
All drift and regulation data are given for max. current output and after warm up time.

Warm up time cold start ..... 60 min.

Line regulation:
±10 % slow. T > 1 min. ..... < 10 ppm
Load regulation:
+10 % resist. change. T > 1min. ... < 10 ppm

PARD (Periodic And Random Deviation), resistive load, f >1 Hz, output voltage, Band Width 100kHz:
PARD..... 1% of V_{out}

Temperature coefficient:
Ambient air ..... < 1 ppm/ºC
Cooling water ..... < 1 ppm/ºC

Ramp time 0 - full scale (Load dependent):
Selectable..... 10 - 1000 sec.

DC output isolation resistance .... > 1 Mohm.
(Without earth leakage detector & cooling water).

Current setting resolution 16 Bit/15ppm
Absolute current calibration .....500 ppm

Current read back resolution...16 Bit/15ppm

Output polarity ..... UNIPOLAR

TEMPERATURE RANGE
OPERATION BELOW 2000M ALTITUDE:
Ambient temperature..... 5 to 40°C
Cooling water temperature..... 15 to 35°C
Relative air humidity..... < 80%

STORAGE temperature ..... 5 to 40°C
Non-Condensing

MAIN COOLING SYSTEM

COOLING WATER PRESSURE:...bar. psi.
Minimum differential pressure..... 3 43
Maximum absolute
(Static pressure)..... 12 174
Test pressure 15 290

COOLING WATER:
Flow (3 bar diff. press).....11 litre/min.
Cooling water fitting ..... 1/2 inch BSP

AC SUPPLY POWER

MAINS VOLTAGE:
XXX V ±10%, three phase, 48 - 62 Hz
4 wire (+ PE) TN system.

MAINS CURRENT:
Per phase ..... <XXAmp.

BNC CONNECTORS ON REAR PANEL

Outputs for read back signals:
Iout 0-10V signal ..... < 2%
Vout 0-10V signal..... < 2%

Inputs:
TTL Trig signal..... 0-5V

OVERALL DIMENSIONS AND WEIGHT

MPS UNIT: (19 Inches wide)..... mm
Height 360
Depth 630

WEIGHT (approx.):.....Kg
Net weight.....90

REMOTE INTERFACE REAR PANEL

Standard serial communication is RS232C

LOCAL IN/OUTPUT INTERFACES

Current setting by UP/DOWN button with a digital indication in three 2-digit groups.

Six line LCD display with four modes. The mode is shown in text and selected by a separate pushbutton.

Pushbutton DC:
DC OUTPUT STATUS mode display:
SET CURRENT in mA
OUTPUT CURRENT in mA
OUTPUT VOLTAGE in Volt
POLARITY
Scale 0-120% of full range

Pushbutton AC:
AC MAIN INPUT STATUS mode display:

PHASE	CURRENT	VOLTAGE
R	AMP	VOLT
S	AMP	VOLT
T	AMP	VOLT

GROUND LEAK CURRENT in mA

Pushbutton INTERLOCK:
INTERLOCK mode display:
DISPLAYING ACTIVE INTL/ STAT
Up to 5 lines are displayed at a time, scrollable with the up/down buttons.

Pushbutton AUX:
AUX mode display:
DELTA TEMPERATURE
OPTIONAL ADC CH1
OPTIONAL ADC CH2

Pushbutton SET:
RAMP & END CURRENT SETUP mode display:
ACTUAL CURRENT in mA
SET END CURRENT in mA
RAMP TIME in 0.1% of MPS max. rated current per second

SET END CURRENT and RAMP TIME are set by UP/DOWN Pushbuttons

Button LEDs and lamps for indication of:
Control Power on..... STAND BY
Main Power on.....ON
Interlock.....INTERLOCK
Local/Remote control mode .....REMOTE

AC Line status .....LINE R
AC Line status .....LINE S
AC Line status .....LINE T

Other pushbuttons:
Start ramping to set current.....RAMP
Reset interlock .....RESET
Up ..... ▲
Down ..... ▼

1.3 Warranty and disclaimer

DANFYSIK A/S warrants the equipment delivered from the company to be free from any defects in materials and workmanship for a period of:

12 Months from the date of installation,

or:

max. 18 months from the date of shipment,

Whichever is shortest?

From the above there are the following exceptions:

Parts not manufactured by DANFYSIK A/S are covered by the warranty from the original manufacturer of the parts.

Within these warranty periods, DANFYSIK A/S will repair or replace any defective parts free of charge either on the customer's site or at our factory at our choice.
Repair work is warranted for 6 months from the date of shipment from the DANFYSIK works.
DANFYSIK A/S will pay or reimburse the lowest freight rate two way charges on any items returned to DANFYSIK A/S or our designated agent/representative provided prior written authorization for such return has been given by DANFYSIK A/S.

This warranty shall not apply to any equipment which our inspection shows to our satisfaction, to have become defective or unworkable due to mishandling, improper maintenance, incorrect use, radiation damage or any other circumstance, not generally acceptable for equipment of a similar type.

DANFYSIK A/S reserves the right on standard products to make changes in design without incurring any obligation to modify previously manufactured units.

The foregoing is the full extent of this warranty and no other warranty is expressed or implied. In no event shall Danfysik be liable for special damage arising from the delivery, late delivery or use of the equipment.

If any fault develops, the following steps should be taken.

Notify DANFYSIK A/S, giving full details of the problems, and include Model-Type and Serial number.

On receipt of these information, DANFYSIK A/S will give you either service information or instructions for shipping.

All shipments of DANFYSIK equipment should be made according to our instructions and shipped in the original or a similar container.

For smaller parts a carton will be sufficient, if the parts are wrapped in plastic or paper and surrounded with at least 10 centimetres of shock-absorbing material.

1.4 General warnings

Please review the following safety precautions to avoid injury and prevent damage to this product or any products connected to it:

To avoid potential hazards, use this product only as specified.

Do not operate with covers or panels removed.

Only qualified personnel should perform service and maintenance procedures!

Install the power supply in accordance with this manual, national wiring rules and local regulations.

This power supply is grounded through the grounding connector at the terminal strip. To avoid electric shock, the grounding connector must be connected to the protective conductor. Before operating the power supply make sure that it is properly grounded.

Do not operate with suspected failures. If damage is suspected to the product, have it inspected by qualified service personnel or contact DANFYSIK A/S.

2 Unpacking and installation

2.1 Receiving the goods

The Shipping container and the Power Supply should be thoroughly inspected for signs of obvious physical damage immediately upon receipt.

All materials in the container should be checked against the enclosed packing list.

DANFYSIK A/S will not be responsible for shortages against the packing list unless notified immediately.

2.2 Instructions for unpacking

The Power Supply is shipped on a wooden pallet enclosed in either reinforced cardboard or wood crate.

To unpack, remove the packing straps and nails if wood crated. The top lid should be removed first.

Warning: Using lifting equipment not rated to carry the weight of the individual units may be dangerous and is consequently forbidden. At least, follow the security rules given by your local authorities. It is dangerous to stand below any burden lifted by a lifting device, e.g., crane or truck. When moving the units, it is dangerous if persons or other obstructions come in the way of the transport equipment.

If the equipment is damaged in any way, a claim should be filed with the shipping agent, and a full report of the damage should be forwarded to DANFYSIK A/S or our local agent/representative immediately.

Upon receipt of this report, you will be issued instructions for the repair, replacement or return shipment.

Please include the Model no, Type no, Serial no, and Order no for the Power Supply on any communication with DANFYSIK A/S or our representative.

Keep the original packing materials and use them in case the equipment needs to be transported again.

2.2.1 When the system 8800 is not to be used any longer

Like almost all other electrical and electronic equipment, the system 8800 Magnet Power Supply system contains materials, which may be harmful to nature if the system is not scrapped or deposited properly. When the system is to be scrapped, it is the responsibility of the owner (user) that all local, national and international rules and recommendations for such scrapping is followed.

2.3 Installation requirements

During installation of the Power Supply, local rules and regulations for electric power and water supplies should be respected and the following conditions and installations should be available.

* A normal, dust free room with a humidity not above 80 % and a room temperature within 5 to 40 centigrade.

* Three-phase Mains voltage, switched and fused, please see the specifications chapter 1.2. for actual figures for this Power Supply.

* Ground connection according to the local authority regulation and the requirements for the equipment.

* Cooling water supply at a temperature within 15 to 35 centigrade.
  Differential pressure: Min. 3 bar.
  Inlet pressure: Max. 12 bar.

* For cooling of our power supplies and magnets we recommend the use of demineralized water or pure water with an electrical conductivity of less than 10µS/cm. This reduces the electrolytic corrosion to a minimum.

During operation, the conductivity of the coolant should be checked regularly. A few days operation with water of poor quality can cause more corrosion damage than during several years operation with good quality water.

Adding pure ethylene glycol to cooling water with no other additives should not affect the electrolytic corrosion. However, it will reduce the thermal conductivity of the coolant. A percentage of 40% glycol reduces the conductivity by approximate 30%. We would recommend not to use more than 25% corresponding to an approximate reduction of thermal conductivity of 17%. It may be necessary to increase the water flow in the power supply.

Warning:
No other additives should be used as they may damage the power supply. DANFYSIK A/S accepts no liability to any damage to the power supply or its environment due to poor water quality, or if any other additive has been used.

2.4 Installation

The installation has 5 phases:

2.4.1. Before the installation: Checking the MPS and the external connections.
2.4.2. Installation of the single MPS unit - General (incl. back panel connections).
2.4.3. Installation of the single MPS unit - Data communication (if required).
2.4.4. Control of the installation.

2.4.1 Before the installation

Before and during installation of the Power Supply, the following points should be checked/carried out.

* Check that the main voltage and frequency matches to the specified and labelled requirements.

* Check that all screw connections carrying heavy current to the output terminals are tightened.

* Check that all plug terminated cables are pushed into final position.

2.4.2 Installation of the single MPS Unit - General

The system 8800 units are normally factory installed in the rack. The only installation to be made to the individual unit is the connection of the computer communication cables/ termination plug. In some cases, there may be a need for changing other parts of the installation, too. The rear panel is described below:

System 8800 MPS unit's rear panel:

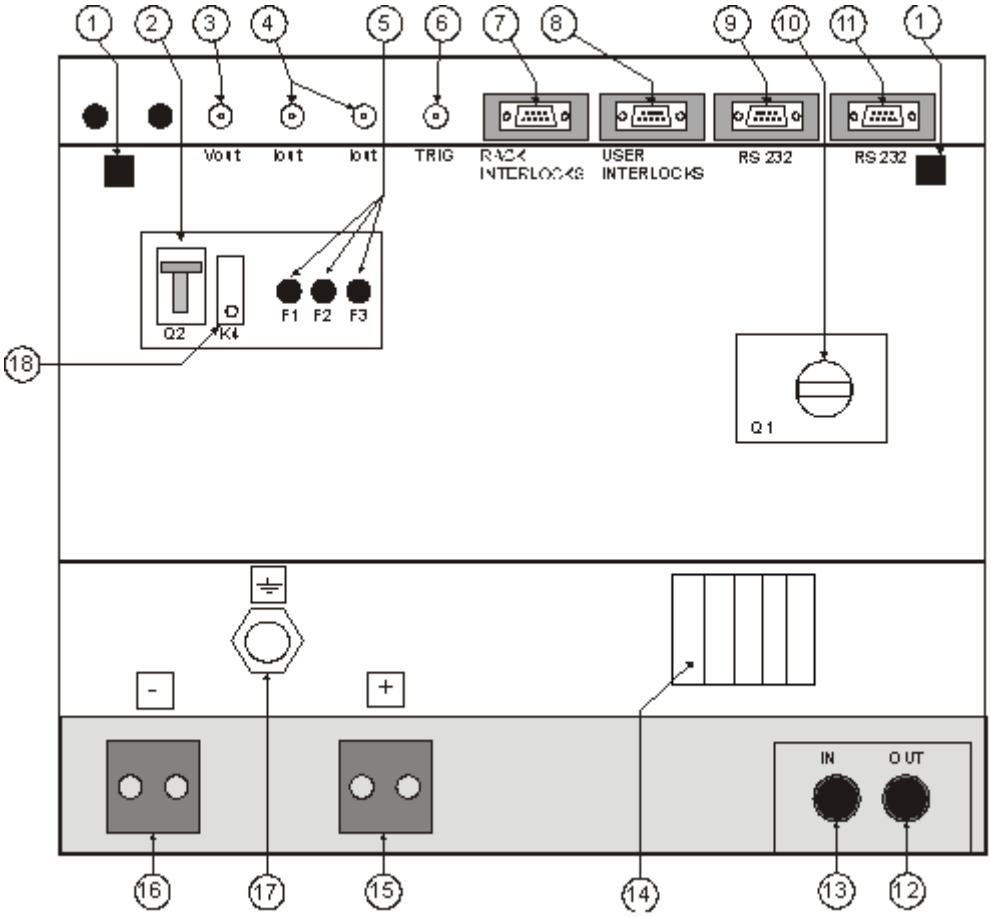


Fig. 2.4-2 The MPS back panel

1. Back panel cover screws (2 pcs.).
2. Control voltage breaker Q2.
3. Voltmeter: output voltage: 0V=0%. 10V=100% of max voltage.
4. Voltmeter: output current: 0V=0%. 10V=100% of max current.
5. Fuses for AC voltmeter and front panel R, S, and T phase LEDs.
6. Trigger input signal. TTL levels. 0->1 = Start.
7. Rack interlocks. Optically isolated 0-24V levels. See next page and Interlocks section 5.2.
8. User interlocks. Optically isolated 0-24V levels. See next page and Interlocks section 5.2.
9. RS232 connector. Pin-for-pin connected to (11). See section 2.4.4.
10. Main on/off switch.
11. RS232 connector. Pin-for-pin connected to (9). See section 2.4.4.
12. Cooling water out.

13. Cooling water in.
14. Power input 1.
15. Magnet current output terminal (+).
16. Magnet current output terminal (-).
17. Ground connection.
18. Power on rely.

Installation of the individual MPS

1. Set the address of the MPS, see 2.4.4.
2. Install the user interlock plug on the USER INTERLOCK connector, see below.
3. Connect the rack interlock plug to the RACK INTERLOCK connector, see below.
4. Install the data communication system, see 2.4.4.

The USER INTERLOCK connector

The user interlocks are controlled by open/closed contacts to 24 V levels.
In section 6.4.3, the pin connections for the User Interlock connector P9 are listed.
An open connection signals an interlock, which will prohibit the system from starting. Enclosed with the power supply is a User Interlock Dummy Plug “DF81083953 USER DUMMY PLUG”, which can be mounted in the USER INTERLOCK back panel connector if no user interlocks are present.
The "user interlocks" are user definable interlocks which may be connected to the system.

* If the magnet is provided with a water flow switch, remove the USER INTERLOCK DUMMY PLUG, and connect the contacts to the USER INTERLOCK connector terminal pin 1 (signal) and pin 9 (common). Remember all unused signal pins 1 to 8 must be connected to pin 9 (common).

* If the magnet is provided with a thermal breaker, remove the USER INTERLOCK DUMMY PLUG, and connect the contacts to the USER INTERLOCK connector terminal pin 2 (signal) and pin 9 (common). Again, remember all unused signal pins 1 to 8 must be connected to pin 9 (common).

The RACK INTERLOCK connector

The rack interlocks are controlled by open/closed contacts to 24 V levels.
In section 6.4.3, the pin connections for the Rack Interlock connector P7 are listed.

Regarding Single Power supply installations, the Rack Dummy Plug “DF81083952 RACK DUMMY PLUG” should be installed in the RACK INTERLOCKS back panel connector, preventing DOOR Interlock and polarity switch Interlock problems.

Concerning rack mounted power supplies with/without polarity switches, the RACK INTERLOCKS connector on the back panel is used for DOOR Interlock and status/control of the polarity switches.

2.4.3 Installation of the single MPS Unit - Data communication

The standard serial communication for this power supply is default set to RS232C.

1. Set the MPS address, see "Setting the MPS's address" below.
2. If necessary, change the Baud-rate, see "Setting the Baud-rate. Bits per byte and parity" below.
3. Install the communication cables, see "Installation of the communication cables" below.

Setting the MPS's address

In order to communicate individually with all the MPS in the system, each MPS must be given an individual address. This is necessary even if all MPS's are identical and should always work identically in parallel, because otherwise the computer cannot receive individual response from each MPS unit. You can address all units simultaneously without response from the units with the order LALL (Listen ALL)

1. Remove the 4 panes lock screws on the *front* panel and pull out the MPS from the rack.
2. Remove the 2 umbraco screws on top of the front panel. Then loosen the 2 umbraco screws on the side panel and fold the front panel down.
3. Remove the 2 Phillips screws from the metal cover above the pc-board and remove the cover. You have now access to the front panel pc-board:

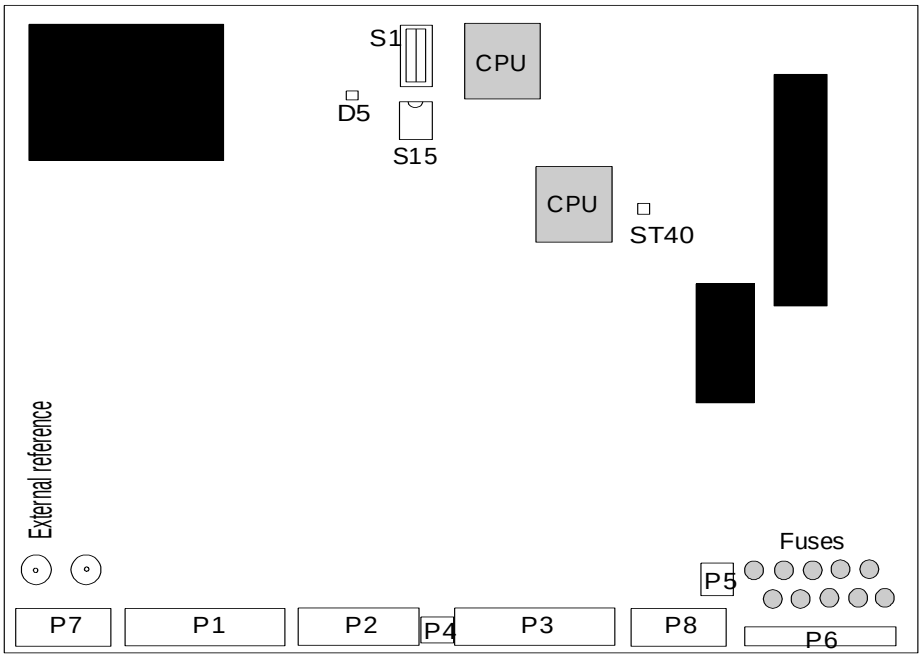


Fig. 2.4-3 The MPS front panel Control board

4. Locate the address DIP-switch S1 (shown on the top of Fig. 2.4-3).
5. Set the binary value of the MPS's address on the DIP-switch:

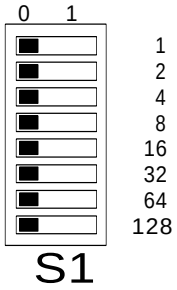


Fig. 2.4-4 The MPS address DIP switch

The address is the sum of the value of the switches, which are set to the 1 position. The address shown is the default address 0.
Please notice: When changing the address settings, the control power has to be turned off due to the fact that address setting is initialized during the control power on sequence.

Setting the Baud-rate. Bits per byte, and parity

The standard serial transfer protocol is 8 bit, no parity and 2 stop bits.

The selection of the Baud-rate is determined by S15, see Fig. 2.4.3 above.
To change the Baud-rate:
- For 57.600 Baud, no straps must be mounted.
- For 38,400 Baud, mount a strap between pin 1 and 8.
- For 19,200 Baud, mount a strap between pin 2 and 7(default factory setting).

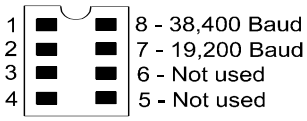


Fig. 2.4-6 The Baud-rate strap field S15
NOTE! The serial lines has no communication control signals.

When all communication settings are completed, remount the MPS, re. point 3 to 1, above.

Installation of the communication cables:

There are 2 connectors for communication cables, See Fig. 2.4-2 (9) and (11): an ingoing connector (from computer) and an outgoing connector (to next MPS unit). They are coupled in parallel, so it does not matter which one you use.

Visual communication test:
To check the communication set-up, you can visually check that you have access to the MPS on-board CPU.
Switch on the MPS, at turn on the MPS should default be in REMOTE mode; From your computer/terminal, address the power supply by sending the string “ADR 000" and press carriage return (provided that the address setting is “000", otherwise you type the actual address setting). The green LED D5 shown on Fig. 2.4-3 should then be lit, indicating that the MPS is addressed, and ready for further commands.

If D5 does not turn on, there is no connection to the MPS, and you must check cables and host computer for errors.

Pin connections RS232C:

The two connectors (9) and (11) are D-SUB Female and Male 9-pin connectors connected in parallel pin for pin.

The direction used in the tables below are:

Rx : Signals received by the MPS from its host.
Tx : Signals transmitted by the MPS to its host.

RS 232C	DB 9
Pin No.	Signal.
2	Tx.
3	Rx.
5	RETURN.

2.4.4 Control of the installation

1. Apply cooling water to the system and check that the system is water tight.
2. Switch the external power on and start the system.
3. Start the system with *manual* control of the magnet current.
4. When the system works OK with manual control, test the system with remote control (if applicable).

3 Operating instructions

This chapter outlines the procedure and precautions for

3.1 Switching on. Necessary before any operation in LOCAL or REMOTE mode.
3.2 Operating in LOCAL mode, incl. use of the display.
3.3 Operating in REMOTE mode, incl. description of all commands available.

3.1 Prepare system 8800 for start

The first time it is recommended to start the Power Supply in LOCAL CONTROL mode.

If not otherwise stated, the numbers below refers to Fig. 3.2-1.

1. Apply external power.
2. If mounted, check the polarity switch setting: 1 = normal polarity; 2 = reverse polarity. The number at the switch relates to the number of the power supply in the rack (1 = bottom).
3. On the rear panel, check that the control voltage breaker Q2 on Fig. 2.4-2 is ON (upper position).
4. On the rear panel, activate the main on/off switch Q1.
   - If the switch falls back to OFF position, points 1-3 above are not fulfilled, or a power line phase is off. Check the LED's LINE R, LINE S and LINE T on the front panel are on.
   - If OK, the red RESET lamp (5), and the green REMOTE (7) and STAND BY (4) lamps will be on. The display will show the DC menu.
5. Reset the interlock: Press INTERLOCK (10) and RESET (5).

3.2 Operating in LOCAL mode - use of the display

3.2.1 The front panel

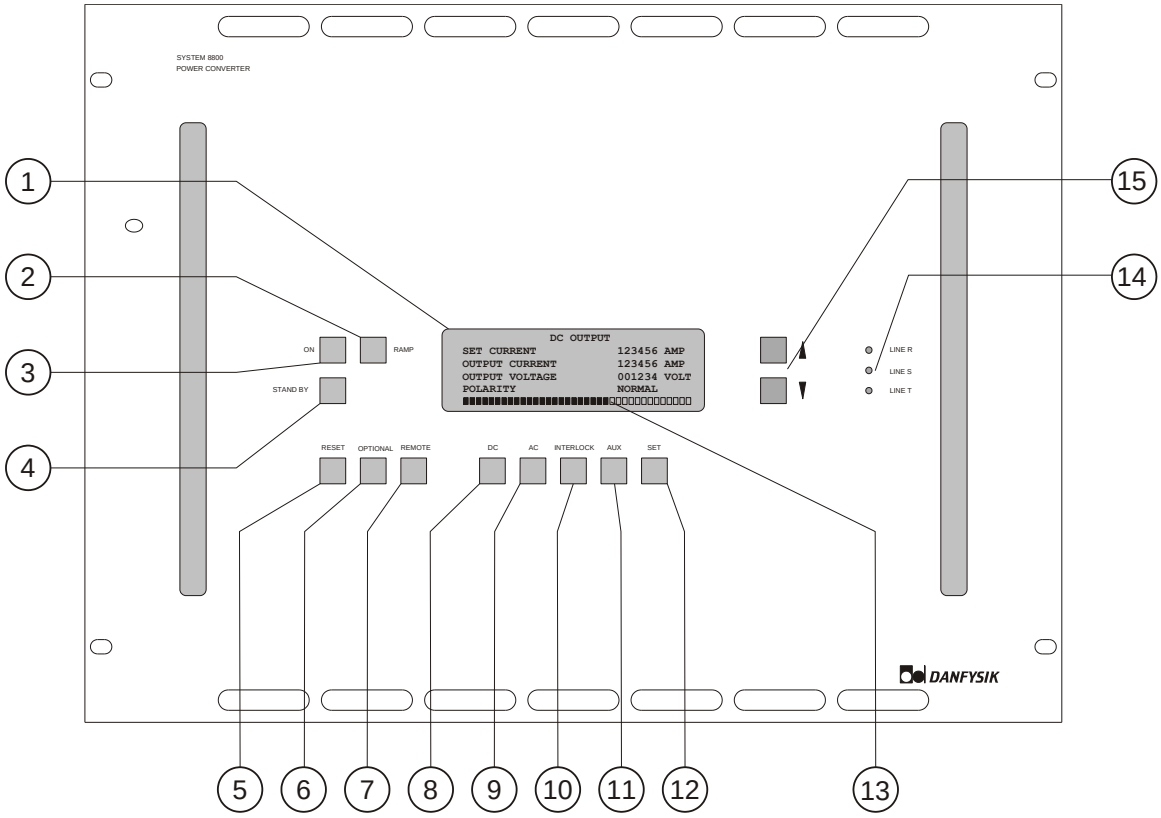


Fig. 3.2-1 The front panel

1. Display.
2. Ramp: Start ramping to the end current wanted. Light in (4) indicates activated function.
3. ON: Output ≥1 Amp. Light indicates activated function.
4. STAND BY. Output: 0 Amp. Light indicates activated function or ramping.
5. RESET: Light indicates one or more interlocks. Press button to reset interlock indication. If this fails, the cause for the interlock still exists.
6. OPTIONAL. This button is disabled.
7. REMOTE mode on/off. Light indicates remote mode.
8. DC: Show DC parameters, see 3.2.3. Default position.
9. AC: Show AC parameters, see 3.2.3.
10. INTERLOCK, see 3.2.3. Press to read the interlock status in the display. In case of several interlocks, Press (15) for more info. Press (5) to reset the interlock(s).
11. AUX: Show AUX parameters, see 3.2.3.
12. SET: Shows ramp and end current parameters, see 3.2.3.
13. Indicator bar on the display.
14. Input AC line status lamps, one per phase.
15. ▲: Up button
    ▼: Down button

3.2.2 Operation in LOCAL mode

Before starting, see 3.1 “Prepare system 8800 for start”.

In the following, if not otherwise stated, the numbers refers to Fig. 3.2-1.

Activate the “Local mode”

1. There are two alternatives to switch to “Local mode”:
   - From a remote terminal: use “Remote control commands”, ADR xxx (or # xx or LALL) and then LOC, see section 3.3.2.
   - On the MPS: press REMOTE (7).

   The REMOTE lamp goes off, indicating “local mode”.

2. Press RESET (5) to remove any “interlock” condition. The red RESET lamp in (5) goes off. If it remains lit, there are still interlock conditions. To solve that, see “Error finds - Interlock lamps on”. If OK: only the green STAND BY lamp is lit.

Set-up in “Local mode“

```
RAMP AND CURRENT SETUP

ACTUAL CURRENT    001000 MA
SET END CURRENT   123456 MA <<--
RAMP TIME         050 PER T.
```

1. Press SET (12) to enter the ramp and current control menu. “<<--” points at the parameter to be set.
   Default when entering this menu, the pointer is ready to set the desired output current value, SET END CURRENT.
2. Set the digits with the UP/DOWN buttons (15). For large numbers an underscore bar indicates which 2-digit group is to be set (the number cannot exceed the limits of the power supply).
3. Press SET (12). The underscore bar moves to the next two digits.
   Set the remaining output current digits by repeating step 2 and 3. The arrow now moves to the
4. RAMP TIME line. The number shows the ramp slope in 0.1 % per second. Example: 052 means 5.2% of *rated max*. current per sec.
5. Set the ramp time using the UP/DOWN buttons (15).
   Press DC (8) to return to the DC menu.

Activate the MPS in “Local mode”

1. Press ON (3). The STAND BY lamp (4) goes off and the ON lamp goes on. The power supply delivers about 1 amp. output current.
2. Press the RAMP button (2). The output current increases to the SET END CURRENT level set in the Local mode setup procedure above. When the ramped reference has reached its final value, the STAND BY lamp (4) goes off. The linear bar (13) shows the setting in % of max. power.
3. When the display shows RDY in the upper right corner, the current is within 200 ppm from the set value. (UND = “undefined”: outside the 200 ppm).
   4. A new output current value can be entered using the procedure described in “Local mode set- up”. When entered, press RAMP (2) to change the output current to the new value.
5. To switch the output current off: press STAND BY (4). The output current ramps down to zero.

3.2.3 The display

To select the display mode press one of the buttons DC, AC, INTERLOCK, AUX or SET. The numbers for the buttons refers to Fig. 3.2-1

DC (8): Status display

```
DC OUTPUT STATUS

SET CURRENT:      123456 MA  RDY
OUTPUT CURRENT:   123456 MA
OUTPUT VOLTAGE:      041 VOLT
POLARITY:         POSITIVE
0       20      40      60       80     100      120
|xxxxxxxxx|xxxxxxxxx|xxxxxxx..|.........|.........|.........|
```

SET CURRENT is the setting in mA. RDY means that the actual OUTPUT CURRENT is within ±200 ppm from the set current. If this is not the case, RDY is replaced by UND (undefined).

OUTPUT VOLTAGE is the voltage across the output terminals of the unit.

The linear bar shows the actual current in % of the max. current accessible from this unit.

AC (9): Status display

```
AC MAIN INPUT STATUS

PHASE      CURRENT       VOLTAGE
R          000 AMP       218 VOLT
S          000 AMP       220 VOLT
T          000 AMP       221 VOLT
GROUND LEAK CURRENT:   0000 MAMP
```

The AC main input status shows the ac input voltage and current to the power supply incl. control power, for each phase R, S and T.

INTERLOCK (10): Status display

```
DISPLAYING ACTIVE INTL/STATUS

X
X
X
X
X
```

"Interlock" means, that the system has been stopped by a fault or for safety reasons.

Pressing the INTERLOCK button displays the current interlock status. To remove an interlock:

Remove the cause of the interlock. The interlock alarm condition remains.
Press RESET to remove the interlock alarm condition in the MPS unit.
Press INTERLOCK to check for more/remaining interlocks.

AUX (12): Status display

```
AUX READOUT MENU DISPLAY

DELTA TEMPERATURE:  -00.1 DEG
OPTIONAL ADC CH1:     000 %
OPTIONAL ADC CH2:     000 %
```

Delta temperature displays the reference cell's temperature deviation from its ideal temperature. It should not exceed ±0.2°C.

The optional ADC channels are not used in this configuration.

SET (12): Command display

```
RAMP END CURRENT SETUP

ACTUAL CURRENT   001000 MA
SET END CURRENT  123456 MA <<--
RAMP TIME        050 PER T.
```

To set the ramp time and end current, see 3.2.2.

3.3 Operating RS 232C I/O

The MPS 8800 uses RS232C as standard serial interface.
The interface link is optically isolated from the magnet current circuit and the power net.

Setting up of the system for serial communication is described in chapter 2.4.4.

The following syntax is used for the commands and answers:
<bell> The control character "bell" (ASCII 7).
<sp> The character "space" (ASCII 32).
<cr> The control character "carriage return" (ASCII 13).
<parameter> The data of the parameter <parameter>.
A The character "A" (ASCII 65)
B The character "B" (ASCII 66)
...

The system accepts upper case letters only. Use of lower case letters will return an error message.

Example:
Syntax: AD<sp><ch><cr>
where <ch> is a number 0-9
means in clear text for ch=9 the transmission of:
"AD 9" (without "") terminated by a carriage return

or in ASCII codes:
65,68,32,57,13.

In a "format":
D is a digit (0-9)

H is a hexadecimal "digit" (0..9,A..F, where A-F is ASCII 65 to 70)
S is a sign (+ or -)
B is a bit (0 or 1).
Example: +09 is the number 9 in the format SDD.

Only the "addressed MPS" can respond to a command. The only exception is when the LALL command is invoked: Here all units will respond to most commands (see the individual command) but no answers are returned.

Errors are reported from the "active MPS", only. This means, that to check for errors, the controlling computer (remote terminal) must address all the relevant MPS addresses regularly. No errors are returned in LALL mode. The error report format is determined by the ERRC or ERRT command. If ERRC or ERRT is *not* activated, the error report is "?<BELL>", only.

The errors listed for each command are the errors which can be returned to the controlling computer for this command. Except in LALL mode, errors are reported even if the command is a "no answer" type.

3.3.1 How to send and receive commands

Before starting, see 3.1 “Prepare system 8800 for start”.

The communication is done by transmitting the commands as ASCII characters terminated by CARRIAGE RETURN (press the Enter key or send ASCII 13).

The termination characters from the Power Supply is CARRIAGE RETURN (ASCII 13).

All ERROR messages include the "?<bell>" where <bell> is the ASCII 7 control character.

NOTE! The serial lines has no communication control signals.

3.3.2 Commands - Index

Command	Description	Remarks
AD xx (x=0-10)	Read value from A/D conv. No. x	x values: see 3.3.3.
ADCV	Reads the actual output current in mA	Format: DDDDDD
ADR xxx or # xx	Call MPS addr. xxx (000-255)	xxx: 000-255. xx: 00-FF. Terminate LALL mode: 2xADR xxx or # xx
ASW	Put WAR and WR in Answer mode	
CMD	Read control mode	REM: Remote mode LOC: Local mode
CMSTATE	Read control mode as text	REMOTE: remote mode (LOCAL: local mode) LOCK: is set to local mode from control panel
ERR*C* / ERR*T*	Return Coded/Text error msg.	See operating instructions 3.3.3
F	Switch magnet power off + reset	
LALL	Listen ALL: Address all MPS units	Multi-MPS system only. Sets MPS answers off.
LOC	Switches the line in command to local operation	
MAX	Read back “max. rated current”	Format: DDDDDD
N	Switch magnet power on	
NASW	Release answer mode	See ASW
NERR	Suppress MPS error mess.	Error gives ?<bell> only. See also ERRC, ERRT
POC	Changes output terminal polarity	Optional
RA	Read actual demanded value of the current in mA	Output: 000000-max rating. Format: DDDDDD.
RAR	Read set end current value in mA	Format: DDDDDD
REM	Switches the line in command to remote operation	
RR	Read slew rate in 0.1% of max setting per sec.	Response: 3 digits
RS	Reset latched interlocks	See operating instructions 3.3.3

S1	Read interlock status	See operating instructions 3.3.3
TS	Trig (start) sequence	Equal to pressing the START button
VER	Returns the programme version	
WA	Write a set value	Sets output current
WAR xxxxxx	Set the magnet current	DDDDDD: Leading zeroes may be omitted
WR xxx	Set slew rate in 0.1% of max setting per sec.	xxx: always 3-digits
?x	Binary read of 18 bit ADC Always 18 bits (0 or 1).	1: Actual current 2: Status (S1) 3: Combined ?1 and ?2 4: ADC at max. output
Ctrl-V	reset the serial communication	

3.3.3 Commands - Details

In the following pages all the commands will be listed in alphabetic order, one command per page.

XX

Address # (write hex)

Syntax: #<sp><address><cr>
where <address> is a number 00 to FF in hexadecimal notation.

Answer: No answer, except errors

Example: Command: # 0C (addressing MPS &h0C = decimal 12)
Syntax: #<sp>0C<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

DATA ERROR The parameter format is incorrect or a non-digit character is found in data field or parameters outside specified. Inspect the settings or correct your parameter, and then rewrite it.

The **ADR** internal command is used to select the actual unit. The previously addressed unit is automatically de-selected.

When the connected units are in **LALL** mode, the first **# xx** command given, following the **LALL** command, will disable this function, only. To address a unit, one more **# xx** command must be send..

Affected commands: LALL

Related commands: ADR xx (write)

AD x

Analog to Digital converter on channel x

Syntax: AD<sp><ch><cr>
where <ch> is the channel number 0 to 10, see the next page in the format D (0-9) or DD (10).

Answer: <val><cr> is a number in the format specified in the table below.

Example: Command: AD 0
Syntax: AD<sp>0<cr>

Answer: 123
Syntax: 123<cr>

Errors: **ILLEGAL REQUEST** Can not activate the addressed MPS unit.

SYNTAX ERROR Wrong syntax, e.g. missing space between the command and the parameter, or wrong number format.

The primary channels giving errors are the internal supply levels, but any of the channels can be programmed to reflect errors if a certain level is passed, both below and above.

Channel <ch>	Value	Units	<val> format
0	R ac voltage	Volt	"DDD"
1	S ac voltage	Volt	"DDD"
2	T ac voltage	Volt	"DDD"
3	R ac current	Amp.	"DDD"
4	S ac current	Amp.	"DDD"
5	T ac current	Amp.	"DDD"
6	Output voltage	Volt	"DDD"
7	Ground leak current	mA	"DDD"
8	Delta temperature, see 3.2.4 the AUX button	Deg.C	"SDDD"
9	AUX CH1	0.1 Volt	"DDD"
10	AUX CH2	0.1 Volt	"DDD"

Where D is a digit 0..9, and S is a sign character (+ or -). The output for channel 9 and 10 measures the voltmeter level outputs for the AUX channel(s) if used.

Affected commands: None.

Related commands: ADCV

ADCV

Analog to Digital ConVerter (18 bit)

Syntax: ADCV<cr>

Answer: <val><cr>
where <val> is a number 000000 to <max rated output current> in the format DDDDDD.

Example: Command: ACDV
Syntax: ACDV<cr>

Answer: 001234
Syntax: 001234<cr>

Errors: **ILLEGAL REQUEST** Can not activate the addressed MPS unit.

SYNTAX ERROR Wrong syntax, e.g. the space between the command and the parameter is missing.

The function reads the output of the 16 bit A/D converter converted to mA, measuring the actual output current from the MPS unit.

Affected commands: None.

Related commands: AD x, RA, ?x.

ADR xx

AdDRess (write)

Syntax: ADR<sp><address><cr>
where <address> is a number 000 to 255 in decimal notation and in the format DDD.

Answer: No answer, except errors

Example: Command: ADR 03
Syntax: ADR<sp>03<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

DATA ERROR The parameter format is incorrect or a non-digit character is found in data field or parameters outside specified. Inspect the settings or correct your parameter, and then rewrite it.

The **ADR** internal command is used to select the actual unit. The previously addressed unit is automatically de-selected.

When the connected units are in **LALL** mode, the first **ADR xxx** command given, following the **LALL** command, will disable this function, only. To address a unit, one more **ADR xxx** command must be send.

Affected commands: LALL.

Related commands: # xx (write hex).

ASW

AnSWer

Syntax: ASW<cr>

Example: Command: ASW
Syntax: ASW<cr>

Answer: No answer, except errors

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **ASW** is used to switch the remote line into an auto-answer mode in which **WR, WAR** and **PO x** will generate an answer. PO x exists only if an automatic polarity switch is installed.

The **ASW** mode is suppressed when the controller is in **LALL** mode.

Affected commands: WAR, WR, PO x.

Related commands: NASW.

CMD

CoMmanD (read control mode)

Syntax: CMD<cr>

Answer: <sp><answer><cr>, where <answer> has 2 possibilities:
REM: the actual unit is in remote control mode.
LOC: the actual unit is in local control mode.

Example: Command: CMD
Syntax: CMD<cr>

Answer: REM
Syntax: <sp>REM<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Local mode has been selected from the units control panel.

The command **CMD** is used to return the control status of the actual unit. It can also be used to detect if anyone had changed the control mode from the control panel, because you will then receive an ILLEGAL REQUEST.

Affected commands: None.

Related commands: CMDSTATE.

CMDSTATE

CoMmanD line STATE (read control mode)

Syntax: CMDSTATE<cr>

Answer: REMOTE the actual unit is in remote control mode.
LOCAL The actual unit is in local control mode.
LOCK the actual unit is locked in local control, and controlled local.

Syntax: <answer><cr>

Example: Command: CMDSTATE
Syntax: CMDSTATE<cr>

Answer: REMOTE
Syntax: REMOTE<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **CMDSTATE** is an extended command similar to **CMD** and is also used to return answer about which unit is the addressed unit.

Affected commands: None.

Related commands: CMD.

ERRC

ERRor in Code (remote line only)

Syntax: ERRC<cr>

Example: Command: ERRC
Syntax: ERRC<cr>

Answer: No answer if there are no errors.
When an error exists or comes on the answer is:
?<BELL><sp><errorcode><cr>
where BELL = ASCII 7

Example: Command: ERRC
Syntax: ERRC<cr>

Answer: ?· 03
Syntax: ?<bell><sp>03<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Cannot activate the addressed MPS unit.

The command **ERRC** is used internally, to put the controller into a mode, in which it will respond any error with the error string followed by two digits, showing which error is encountered.

This mode is normally chosen, when the controller(s) is (are) connected to a host computer, which is able to decode the error message.

CODE
01 COMMAND ERROR
02 DATA PARM ERROR
03 DATA SIZE ERROR
04 ILLEGAL COMMAND
05 RAMP RUNNING
06 STATUS QUO – NO CHANGE
07 CHANGE IN PROGRESS-COMMAND DELETE
08 STACK IS RUNNING
09 TO MANY PARMS ERROR
10 MISSING PARMS ERROR
11 PARM OVERFLOW ERROR
12 PSU ERROR
13 NOT READY ERROR
14 SYNTAX ERROR
15 NOT USED ERROR
16 NOT IMPLEMENTED

Affected commands: None.

Related commands: ERRT, NERR

ERRT

ERRor in Text **(remote line only)**

Syntax: ERRT<cr>

Example: Command: ERRT
Syntax: ERRT<cr>

Answer: No answer if there are no errors.
If/when an error exists or comes on the answer is:
?<bell><sp><error text><cr>

Example: Command: ERRT
Syntax: ERRT<cr>

Answer: ?· <Error text>
Syntax: ?<bell><sp>DATA CONTENTS<cr>

Errors:	**COMMAND ERROR**	Spelling error in a command
	SYNTAX ERROR	Wrong syntax, e.g., wrong number of digits.
	ILLEGAL REQUEST	Can not activate the addressed MPS unit.

The command **ERRT** is used internally, to put the controller into a mode, in which it will respond any error with a message containing an error string followed by the error text.

This mode is normally chosen, when the controller(s) is (are) connected to a host computer, which is *not* able to decode the error message.

CODE	
01	COMMAND ERROR
02	DATA PARM ERROR
03	DATA SIZE ERROR
04	ILLEGAL COMMAND
05	RAMP RUNNING
06	STATUS QUO – NO CHANGE
07	CHANGE IN PROGRESS-COMMAND DELETE
08	STACK IS RUNNING
09	TO MANY PARMS ERROR
10	MISSING PARMS ERROR
11	PARM OVERFLOW ERROR
12	PSU ERROR
13	NOT READY ERROR
14	SYNTAX ERROR
15	NOT USED ERROR
16	NOT IMPLEMENTED

Affected commands: None.

Related commands: ERRC, NERR.

F

ofF

Syntax: F<cr>

Answer: No answer, except errors.

Example: Command: F
Syntax: F<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **F** is equivalent to pressing the STAND BY button (5) on the front panel, see Fig. 3.2-1.

All settings of ramp and end current is reset.
Ramp is set to 5%/sec. of max rated current and end current is set to 1 Amp.

Affected commands: N, WAR, WR.

Related commands: N, GOFF.

F1

ofF Auxiliary

Syntax: F1'er’

Answer: No answer, except errors

Example: Command: F1
Syntax: F1'er’

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST means that line-in-command is wrong.

The **F1** command is used to switch-off the Auxiliary line output port

Affected commands: None.

Related commands: F, N, N1

LALL

Listen ALL

Syntax: LALL<cr>

Answer: No answers!

Example: Command: LALL
Syntax: LALL<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

The command **LALL** is used to put all the controllers connected into a pseudo-addressed mode. This means that all controllers will respond to all setup (e.g., WAR) and action (e.g. o**N**) commands. No answers will be available.

The only way to disable the **LALL** mode is by using the **ADR** command either as a setup of a new address or to read last addressed controller. Remark that in **ADR** read, the first access will not give any response, in which way you will have to repeat the command to get an answer.

Affected commands: ADR xxx, # xx, ASW.

Related commands: ADR xxx, # xx.

LOC

LOCal (Line)

Syntax: LOC<cr>

Answer: No answers, except errors

Example: Command: LOC
Syntax: LOC<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

The command **LOC** is used to switch the line-in-command to the local line.

Affected commands: None

Related commands: REM.

MAX

Read MAX current setting

Syntax: MAX<cr>

Answer: The maximum rated current for the MPS unit in mA in the DDDDDD format.
Syntax: <max><cr>

Example: Command: MAX
Syntax: MAX<cr>

Answer: 400000
Syntax: 400000<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **MAX** reads the maximum rated current of the addressed MPS.

Affected commands: None.

Related commands: None.

N

oN

Syntax: N<cr>

Answer: No answer, except errors

Example: Command: N
Syntax: N<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **N** is equivalent to pressing the ON button (4) on the front panel, see Fig. 3.2-1. It sets the magnet current to about 1 Amp.

Note: This command can *not* be used in **LALL** mode.

Affected commands: F.

Related commands: None.

NASW

No AnSWer

Syntax: NASW<cr>

Answer: No answer, except errors

Example: Command: NASW
Syntax: NASW<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **NASW** is used to cancel the auto-answer mode, in which some setup commands generate an answer.

Affected commands: None.

Related commands: ASW

NERR

No ERRor

Syntax: NERR<cr>

Answer: No answer, except errors

Example: Command: NERR
Syntax: NERR<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **NERR** puts the controller into a mode, in which it responds to any error with the error string ?<bell> only, without any identification of the error.
This mode is chosen, if you only want to be kept informed that an error has occurred.

Affected commands: None.

Related commands: ERRC, ERRT

POC

POlarity Change

Syntax: POC<cr>

Answer: No answer, except errors

Example: Command: POC
Syntax: POC<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The **POC** command is equivalent to pressing the POLARITY button (7) on the front panel, See Fig. 3.2-1. The command changes the polarity of the output voltage.

Note: If there is no motorized polarity switch build-in, this command is not available.

Affected commands: None.

Related commands: None.

RA

Read Actual demanded current in mA

Syntax: RA<cr>

Answer: <ra><cr>
where <ra> is a digital value 000000 to <max rated current in mA> in the format DDDDDD.

Example: Command: RA
Syntax: RA<cr>

Answer: 004800
Syntax: 004800<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **RA** reads the actual DAC setting in mA. It is used e.g. during ramping.

Affected commands: None.

Related commands: RAR.

RAR

Read end current in mA

Syntax: RAR<cr>

Answer: <rar><cr>
where <rar> is a digital value 000000 to <max rated current> in the format DDDDDD.

Example: Command: RAR
Syntax: RAR<cr>

Answer: 2560000
Syntax: 2560000<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **RAR** reads the set end current value in mA.

Affected commands: None.

Related commands: WA, RA

REM

REMote (line)

Syntax: REM<cr>

Answer: No answer, except errors

Example: REM
Syntax: REM<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

The REM command is used to switch the line-in-command to the remote operation.

Affected commands: None.

Related commands: LOC

RR

Read Ramping slope in 0.1%/sec. of max rated current

Syntax: RR<cr>

Answer: <ramp><cr>
where <ramp> is a digital value 000 to 100 in the format DDD.

Example: Command: RR
Syntax: RR<cr>

Answer: 480
Syntax: 480<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **RR** is used to read the ramping slope, expressed in 0.1%/sec. of max rated current.

Affected commands: None.

Related commands: RA, RR, RAR.

RS

ReSet

Command: RS<cr>

Answer: No answer, except errors

Example: Command: RS
Syntax: RS<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **RS** clears all interlocks, if the reasons for the interlocks are removed. It is equivalent to pressing the RESET button (6) on Fig. 3.2-1. All settings are left un-affected.

Affected commands: None.

Related commands: None.

S1

Status 1

Syntax: S1<cr>

Answer: <status>
Syntax: <status><cr>
where <status> consists of 32 signs . or ! , each showing the status of a specific function, including all interlocks. A list is shown on the next page.

Example: Command: S1
Syntax: S1<cr>

Answer: ..!.....!.....!....!.
Syntax: ..!.....!.....!....!.<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **S1** is used to return answer about the internal status. The returned status line consists of a mixture of interlocks, polarity and on/off status.

The status command can be used in several ways to check polarity (+,- or none), to indicate type of interlocks and on/off status.
In some versions some spare bits can be assigned special functions.
Each sign is specified separately elsewhere in the user manual.

Affected commands: None.

Related commands: None.

S1 Continued

A typical example could be as follows:

". . ! . . . . . ! . . . . . ! . . . . ! ."
0 Character. 31st Character.

The interpretation of the individual characters, when the exclamation mark (!) is shown, is as follows:

Char. No.	Type	Contents
0	Spare	
1	Interlock	User interlock 1
2	Interlock	User interlock 2
3	Interlock	User interlock 3
4	Interlock	User interlock 4
5	Interlock	User interlock 5
6	Interlock	User interlock 6
7	Interlock	Freewheel diode overtemp.
8	Interlock	Low water Flow
9	Interlock	Door open
10	Status	Positive polarity. Note 1
11	Status	Negative polarity. Note 1
12	Spare	
13	Spare	
14	Spare	
15	Spare	
16	Interlock	Diode heatsink overtemp.
17	Interlock	Chassis overtemp.
18	Interlock	IGBT heatsink overtemp.
19	Interlock	H.F. diode overtemp.
20	Interlock	Switch reg. DCCT failure
21	Interlock	Switch regulator supply failure
22	Interlock	IGBT driver failure
23	Spare	
24	Spare	
25	Warning	AC under voltage
26	Spare	
27	Warning	Excessive current ripple
28	Warning	Ground leak current
29	Interlock	Over current
30	Status	Main power is on
31	Status	Ready

Note 1: Neutral: 10 and 11 is both "!", and this is an interlock.
Not existing: 10 and 11 is ".".

TS

Trig Start sequence

Syntax: TS<cr>

Answer: No answer except errors

Example: Command: TS
Syntax: TS<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **TS** is used to start the ramping function. It is equivalent to pressing the START button on the MPS panel, see (3) on Fig. 3.2-1.

Affected commands: None.

Related commands: STOP

VER

Software VERsion

Syntax: VER<cr>

Answer: *<sp><type><sp><version>sp>*<cr>
where <type> is the text "DANFYSIK A/S SYSTEM 8800" and <version> is 2 letters, see the example below.

Example: Command: VER
Syntax: VER<cr>

Answer: * DANFYSIK A/S SYSTEM 8800 SB *
Syntax: *<sp>DANFYSIK<sp>A/S<sp>SYSTEM<sp>8800>sp>SB<sp>* <cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **VER** is used to read the software version of the MPS control board.

Affected commands: None.

Related commands: None

WA

Write DAC in ppm

Syntax: WA<cr>

Answer: No answer, except errors.

Example: Command: WA 10000
Syntax: WA ‘sp’10000’cr’
This means 10000 ppm

Answer: No answer except errors

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command WA is used to write an PPM value between 0 and 999999 to the regulation module DAC for output set current.

Affected commands: None.

Related commands: RA

WAR xxxxxx

Write end current in mA

Command: WA<sp><endcurr><cr>

<endcurr>: digits 1000 to <max rated current> in mA. Format DDDDDD. The value must be >1000 mA.

Remark: Leading zeroes can be omitted.

Answer: <endcurr> In ASW mode: digits 000000 to 999999. Format DDDDDD.
Else: No answer.

Example: Command: WAR 004800
Syntax: WAR<sp>004800<cr>
Answer: 004800
Syntax: 004800<sp><cr>

Errors:	**SYNTAX ERROR**	Missing space between the command and parameters or wrong syntax.
	DATA ERROR	Incorrect parameter format, or a non-digit character found in data field or parameters outside specified. In case of this error you should inspect the settings or correct your parameter, and then rewrite it.
	ILLEGAL REQUEST	Indicates that you are in a wrong command mode. Change REMote or LOCal.
	CHANGE IN PROGRESS	Indicates that the controller are doing an internal sequence as polar change. While this is running, it is not allowed to make a new setup.

The command **WAR** is used to write the end current in mA, value between 0 and the max rated current for the MPS unit.

Affected commands: None.

Related commands: WR, STOP, (ASW)..

WR xxx

Write Ramp

Syntax: WR<sp><ramp><cr>
where <ramp> is the ramp slope in 0.1% per second. Format DDD.

Answer: <ramp> In ASW mode: digits 001 to 100. Format DDD.
Else: No answer.

Example: Command: WR 100
Syntax: WR<sp>100<cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **WR** is used to determine the slope (rate) for the ramping function.

Affected commands: None.

Related commands: STOP (ASW)

?x

Binary read ADC data etc.

Syntax: ?<par><cr>

Answer: The answer depends of the value of <par> which can be 1, 2, 3 or 4.
Syntax: <result><cr>
where <result> can be:

?1 24 bit for AD converter value.
?2 32 bit for S1.
?3 Combination of ?1 and ?2.
?4 24 bit.

1 Actual current as AD converter out (=ADCV)
2 Status: as S1, but with "." as 0 and "!" as 1.
3 As 1 and 2 combined: first 18 bit for actual current and then 18 bit for status. The format for <result> is <actual current><S1>
4 The AD converter value of the MPS rated maximum current.

Example: Syntax: ?1<cr>
Command: ?1

Answer: 001010001010100101011010
Syntax: 001010001010100101011010<cr>

Command: ?3
Syntax: ?3<cr>

Answer: 000110000001010001001000000100000000000000010000101011000
Syntax: 000110000001010001001000000100000000000000010000101011000 <cr>

Errors: **SYNTAX ERROR** Wrong syntax.

ILLEGAL REQUEST Can not activate the addressed MPS unit.

The command **?x** is used for fast reading of the essential data making the terminal computer able to calculate display the output current and status in its own format without losing anything on rounding.

Affected commands: None.

Related commands: ADCV, S1, RA

4 Maintenance

4.1 Introduction

Servicing DANFYSIK Magnet Power Supply should be attempted only by trained and qualified personal.

Dangerous voltages capable of causing loss of life are present inside this power supply. Use extreme caution when accessing, handling, testing and adjusting.

4.2 Preventive maintenance

In normal operating environment, perform the following tasks at one-year intervals:

-Clean all fan protection grilles.

-Vacuum the openings in the cabinet and all heat sinks mounted on printed circuit boards to ensure a normal flow of cooling air.

-Check that all cables and bus bars carrying heavy current are tightened.

-Check that the water-flow switch works correctly.

In dusty or dirty environments, the above-mentioned points should be performed more often.

-Inspect visually the power supply for components that have been overheated or other suspicious signs.

In high radiation environment, performance of following tasks are recommended at one-year intervals:

-Replace all IC’s mounted in sockets on the printed circuit boards.

-Printed circuit boards with IC’s mounted without sockets should be replaced.

4.3 Adjustment and calibration

This power supply do not need any regular adjustment or calibration.

5 Trouble shooting

The syntax for trouble shooting hints is:

SYMPTOM:
-Check #1:
> Details (new check or failure if no other failure is found by the checks)

-Check #2:
> Details (new check or failure if no other failure is found by the checks)

In case of a failure...

1. Check if any of the LINE R, S or T lamps are off. If so, see section 5.1 general failures.

2 If possible, read the interlock list, either on the display (see section 3.2.1, point 10, and 3.2.4) or by using the S1 command (see section 3.3). If there are interlocks, see the interlock(s) in section 5.2.

3. Check the general failures list in section 5.1

4. Observe which LEDs are illuminated on the Control panel or the Switch module, see Fig. 5.1-1, and section 6.1.2.

5. If you cannot remedy the failure yourself, call your Danfysik service.

5.1 General failures

General failures are failures, where the system is *not* able to show an interlock.

THE R, S OR T LAMP IS OFF
- Check the external breakers and fuses.
- Check Q1 on the back panel, see section 3.1

Q2 FALLS BACK TO OFF POSITION WHEN ACTIVATED
- Check if EMERGENCY STOP has been activated.

NO RESPONSE ON CONTROL PANEL:
- Check the internal fuses:
  > Get access to the fuses:
  1. Remove the 4 panel lock screws on the front panel and pull out the MPS.
  2. Remove the 2 umbraco screws on top of the front panel, loosen the 2 screws on the side panel, and fold the front panel down.

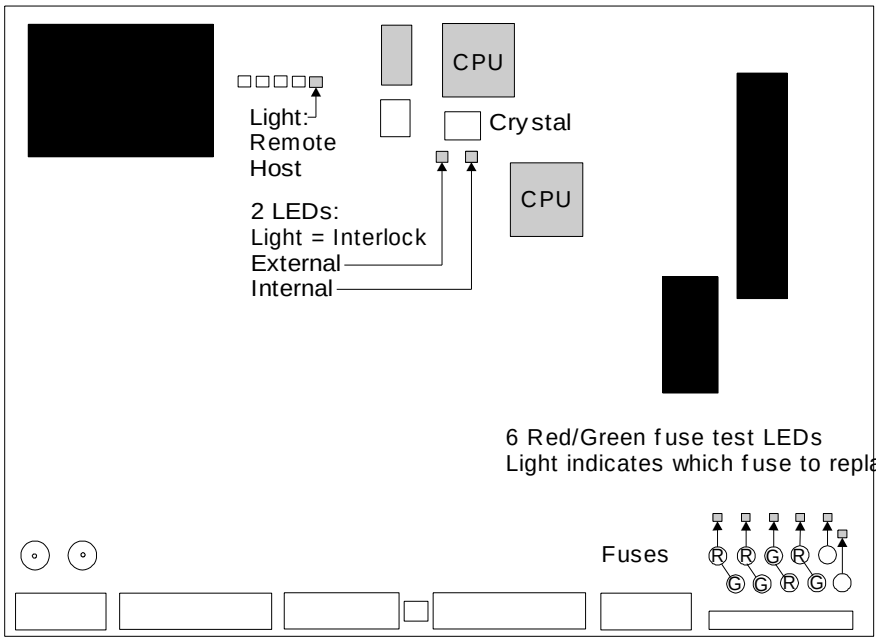


Fig. 5.1-1 The control board with LED positions.

  3. In case any of the 6 fuse-test LEDs is not red+green, switch off all power and replace its related fuse:
     - A red R or green G light, See Fig. 5.1-1, marks that the related R or G marked fuse is blown.
     - For unmarked fuses, both red+green light means OK, no light means fuse blown.
  4. Remount the front panel. If the fuse blows repeatedly: Contact your Danfysik service.
     -Check the cable connections to the control board and the control panel.

  > Failure on the Control board or the control panel.

WILL NOT GIVE OUTPUT POWER:

- Check for interlocks.
- Check that the polarity switch is not in position Neutral (0).
- Check the connections to the magnet power output terminals and the polarity switch.
  > Failure on the Control board.

NONE OR BAD REGULATION:

- Check the AC mains voltage, see section 3.2.3, the AC button.
  > The AC voltage on all phases should be within the limits specified in section 1.2 - AC POWER SUPPLY.

- Check the temperature readout, see section 3.2.3, the AUX button :
  > Delta temperature is more than +/- 0.5°C offset (it should approach +/- 0.2°C): Failure on the regulation board.

- Check the ambient temperature, see section 1.2 - TEMPERATURE RANGE.:
  > Temperature too high:
  Ambient temperature must come down.

- Check the cooling water temperature:
  > Temperature low:
  Risk of dew in the electronics. Raise the cooling water temperature .

- Check the OUTPUT VOLTAGE, see 3.2.3 - the DC button. For max permissible outputvoltage, see 1.2 - DC OUTPUT RATINGS.

5.2 Interlocks

The interlock system is also described in section 6.3.

When the interlocks are handled, press the front panel RESET pushbutton.
Then press INTERLOCK pushbutton to see if there are more problems.
The difference between interlocks and warnings is that an interlock switches the power supply off, and it cannot be turned on before the interlock is removed. A Warning means that a level has passed the preset trig-level, but the power supply keeps on running!

DOOR OPEN
- Check the rear panel door of the cabinet.
  > If it was closed, check the 3 panel door contacts one by one.

NEUTRAL POLARITY
- Check the polarity switch(es).

OVER CURRENT
- Check the connections to the DCCT (current measurement transducer) head.
  > Failure on the control module (see 6.1.1) or the SMC module (see 6.1.2).

LOW WATER FLOW
- Check the water cooling system.

IGBT HEATSINK OVERTEMP
- Check the cooling water temperature, see section 1.2 - TEMPERATURE RANGE.:
  > Too high temperature :
    The cooling water temperature must come down.
- Check the internal water cooling.

H.F. DIODE HEATSINK OVERTEMP
- Check the cooling water temperature, see section 1.2 - TEMPERATURE RANGE.:
  > Too high temperature :
    The cooling water temperature must come down.
- Check the internal water cooling.

FREEWHEEL DIODE TEMP
- Check the temperature on the free-wheel diode and the output busbar

SWITCH REG. SUPPLY FAILURE
- Check the voltage on the SMPC Regulation board, see section 6.1.2.

SWITCH REG. DCCT FAILURE
- Check the connection between the Control board and the DCCT head.

IGBT OR DRIVER FAILURE

- Check the connection between the SMPC Regulation board and the IGBT Driver boards.
- Check the connection between the IGBT Driver boards and the IGBT modules.
  > Failure in the IGBT module
- Check that the 4 green LED's (2 on each IGBT Driver Module) are on.
  > Failure in the IGBT Driver/IGBT Module

CHASSIS OVERTEMP

- Check the ambient temperature, see section 1.2 - TEMPERATURE RANGE.:
  > Too high temperature :
  The ambient temperature must come down.
- Check the cooling water temperature, see section 1.2 - TEMPERATURE RANGE.:
  > Too high temperature :
  The cooling water temperature must come down.

USER 1-6 or Magnet Water/Magnet Temperature:

- Check the connections and ext. interlocks according to section 2.4.3.

AC UNDERVOLTAGE (Warning)

- Check the AC input voltage. You cannot turn the power supply on before this warning is removed

GROUND LEAK CURRENT (Warning)

- Check the load connected to the power supply, current flowing from the output terminals to ground has passed the preset trig-level.

5.3 Repair and calibration

This section gives instructions and informations on how to handle repairings, modification and calibration of the MPS883 power supply.

5.3.1 Repair and board calibration

As shown in section 6, the power supply is divided into three major sections: the AC Module, the Switch Mode power Converter Module and the Control Module.
Each section is mounted in its own box, and each box is interfaced with connectors, thereby each section is very easy to remove for repair, or to replace with a spare.
This keeps the down time low, and qualified personnel should not see problems in replacement of any main section. In section 6 the access to each module and the related components is described.

WARNING: Before disassembly of the power supply, please note: Only trained and qualified personnel should be allowed to repair or service the power supply. Dangerous voltages capable of causing loss of life are present inside this power supply; Therefore show extreme caution when accessing, handling and servicing the power supply.

Regarding spares, it is possible that some spare Printed Circuit Boards has to be calibrated on the specific power supply. In this case please consult the test report for more descriptions and procedures, or contact DANFYSIK A/S or your local agent/- representative.

Once again concerning repair of individual modules or components, only trained and qualified personnel should perform this task. If you have any problems or questions regarding repairs please contact your DANFYSIK representative or DANFYSIK A/S.

5.3.2 Setup key calibration of MPS883

Warning:

Entering the calibration menu of the MPS883 power supply might create unintentional damage, because the factory set calibration data might be lost! Therefore, remove the Set-up key, if mounted, from the front plate of the power supply, and keep it on a safe place to avoid unintended use.
Do only enter the calibration menu if you are 100% certain about what you do, otherwise contact your DANFYSIK service before starting.

Please be aware that the calibration procedure is an absolute calibration, ie. the power supply must be calibrated when delivering rated output power. Furthermore notice how to save the calibration data when leaving the menus.

If the calibration key is not mounted on the front panel, short-circuit the 2 pins at P5 on Figure 1, by doing this you enter the calibration menu.

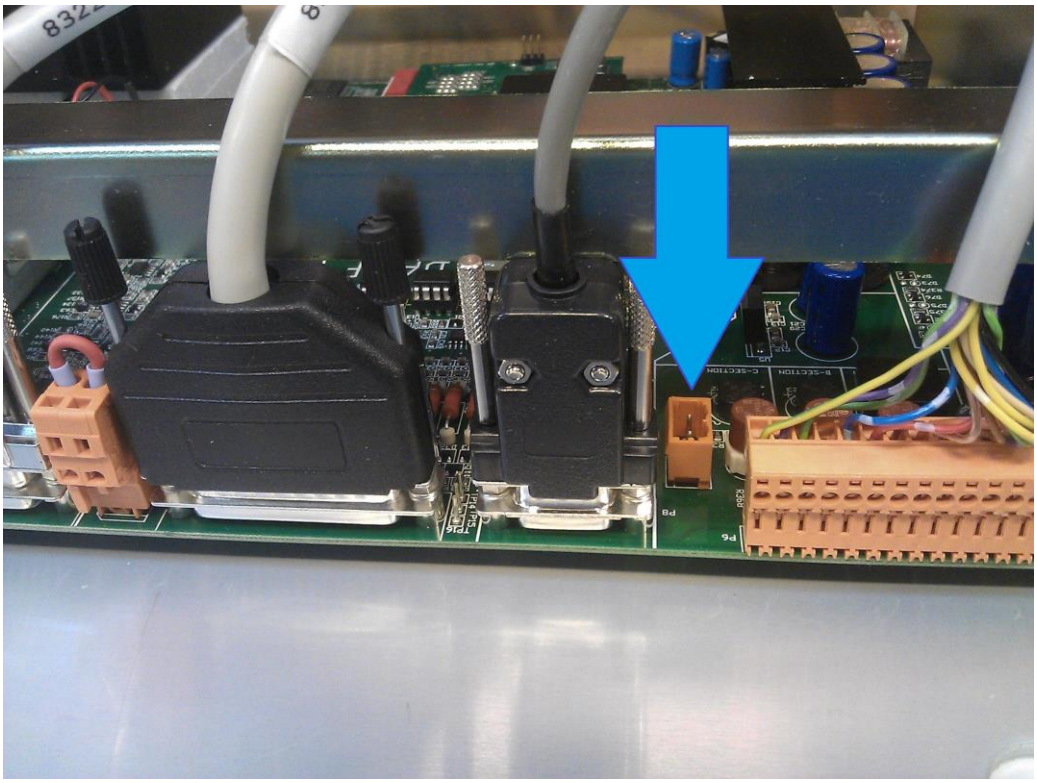


Figure 1: P5

If not otherwise stated, the pushbuttons mentioned below refer to Fig. 3.2-1

Calibration Procedure:

- Main is ON and the MPS is `ON´
- Turn SET UP key or short-circuit P5.

Display:

SERVICE SWITCH ACTIVATED
DC-AC AUX KEYS ACTS DIFFERENT
USE: DC-KEY TO: DC-SETUP
USE: AC-KEY TO: AC-SETUP
USE: AUX-KEY TO: AUX-SETUP
REMAINING KEYS WORKS AS USUAL

Explanation:

DC-SETUP: To make an absolute calibration on DC output data.

AC-SETUP: To make an absolute calibration on AC input data.

AUX-SETUP: To select Reference source and feedback source for current loop.

DC-SETUP menu.

- Push the DC pushbutton.

  Display :

  DC SETUP MENU PICTURE
  AFTER ANY SETUP HIT DC-KEY TO
  GO TO NEXT STEP
  >SET OUTPUT TO 100%: 000000
  SET 100% OUTPUT TO: 215000 MA
  SET MEASURED VOLTAGE: 025 V

  Explanation:

SET OUTPUT TO 100%: Set DAC value/current reference for rated output current (absolute calibration of current).

SET 100% OUTPUT TO: Set measured value of current at output at 100% (absolute calibration of current read back).

SET MEASURED VOLTAGE: Set value of measured voltage on output at 100% (absolute calibration of voltage read back).

The select line is indicated by > and _ indicate which figure you can change.

- Push the UP or DOWN button to change value.

- Push the DC button to change _ figure in display, push 3 times at DC button and > go to next step in the menu.

DC-Calibration of MPS:

- Measure the output current to load by an external measuring system/instrument.

- Select menu **SET OUTPUT TO 100%.**
  Push UP or DOWN button until rated output current is flowing.

- Select menu **SET 100% OUTPUT TO.**
  Push UP or DOWN button until the display shows the measured output current

- Measure the output voltage by an external measuring instrument/DVM.

- **Select menu SET MEASURED VOLTAGE.**
  Push UP or DOWN button until the display shows the measured output voltage.

- To store data in the CPU, after last step push DC button > go to first step in the menu and setup data for this menu are stored in CPU on the control bard.

- AC-SETUP menu

- Push the AC push-button

Display :

AC SETUP MENU PICTURE
AFTER ANY SETUP HIT DC-KEY TO
GO TO NEXT STEP
>SET MEASURED CURRENT: 100 AMP
SET MEASURED VOLTAGE: 230 VOLT
GND-LEAK TRIG LEVEL: 412 MAMP

Explanation:

SET MEASURED CURRENT: Set value of measured current on AC input line (absolute calibration of AC current read back).

SET MEASURED VOLTAGE: Set value of measured voltage on AC input line (absolute calibration of AC voltage read back).

GND-LEAK TRIG LEVEL: Set value for ground-leak current trig level

The select line is indicated by > and _ indicate which figure you can change

- Push the UP or DOWN button to change value.

- Push the AC button to change step in the menu.

AC Calibration of MPS:

- Measure the AC input current to the MPS by an external measuring instrument.

- Select menu **SET MEASURED CURRENT.**
  Push UP or DOWN button until the display shows the measured AC input current.

- Measure the AC input voltage with an external measuring instrument/DVM.

- Select menu **SET MEASURED VOLTAGE.**
  Push UP or DOWN button until the display shows the measured AC input voltage.

- Select menu **GND-LEAK TRIG LEVEL.**
  Push UP or DOWN button until the display show the level for ground-leak warning.

- To store data in the CPU, after last step push AC button > go to first step in the menu and setup data for this menu are stored in CPU on the control board.

- AUX menu.

- Push the AUX push-button

Display:

REF/F.B. SOURCE SETUP PICTURE
AFTER ANY SETUP HIT AUX-KEY TO
GO TO THE NEXT STEP
>SELECT REF. SOURCE: INTERNAL
SELECT F.B. SOURCE: INTERNAL
USE SCROLL-UP KEY TO TOGGLE

Explanation:

SELECT REF. SOURCE:	Select reference source for current loop.
INTERNAL:	Current reference from DAC on the control board is used.
EXTERNAL:	Current reference from BNC plug on control board, see Fig. 6.4.1-1. An external 0-10V voltage source controls output current from 0 to rated output current. **NOTICE:** The accuracy and stability of the external current reference source influences directly on the stability of the output current of this power supply.
SELECT F.B. SOURCE:	Select source for feed back unit used in current loop.
INTERNAL:	DCCT inside the power supply.
EXTERNAL:	External unit to measure current/field in the magnet. **WARNING:** By using an external feedback element you open the high gain (more than 60dB!) control loop, and removes several phase compensating components. It is strongly recommended NOT to use this feature unless you are extremely carefull to assure a stable control loop

AUX-Calibration of MPS:

- Push the UP button to change INT / EXT.
- Push the AUX button to change step in the menu.
- Push the UP button to change INT / EXT
- To store data in the CPU, after last step push AUX button > go to first step in the menu and setup data for this menu are stored in CPU on the control bard.

Exit of calibration menu:

When the calibration is completed, Turn the SET UP key back to normal position and remove the key!If key not mounted, remove the short-circuit at P4!

6 Technical description

NOTE: Change notes are placed in section 9.

The power supply consists of three modules (Fig. 6-1):
- Control module placed at the front of the MPS, see Fig. 6-2 and section 6.1.1.
- Switch Mode power Converter module placed in the centre of the MPS, see Fig. 6-2 and section 6.1.2.
- AC input module, placed at the rear of the MPS, see Fig. 6-2 and section 6.1.3.

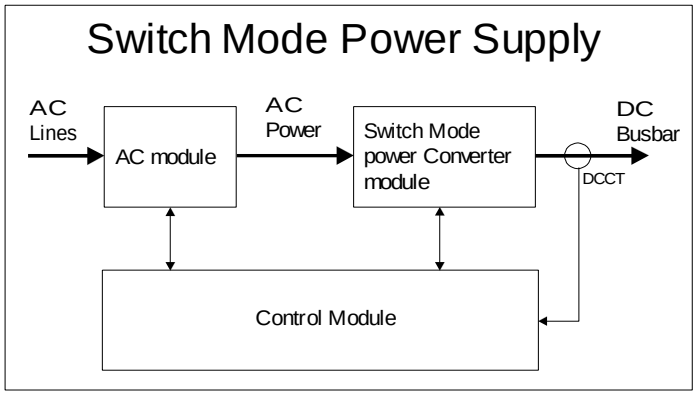


Fig. 6-1 The MPS883 Switch Mode Power Supply.

The modules are mounted in a U shaped chassis, 19" wide, 350 mm high and 600 mm deep, mounted in the cabinet on telescopic rails. This allows for a very quick replacement of all modules, thus keeping down the service-time and total downtime.

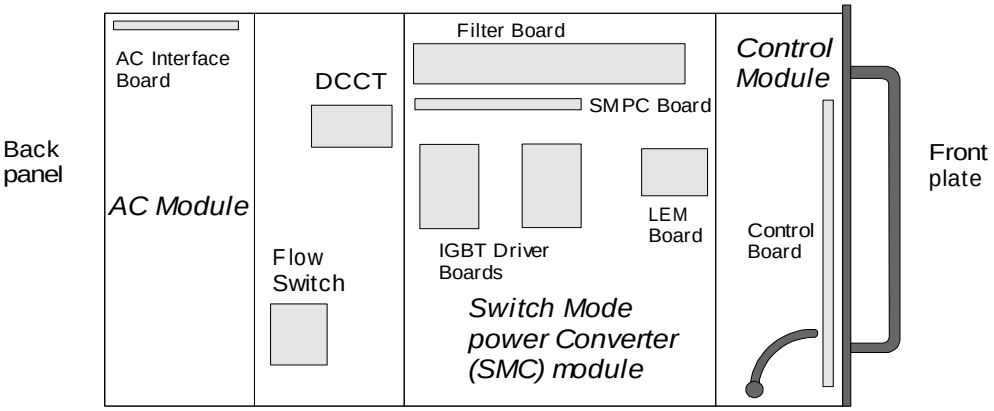


Fig. 6-2 The MPS unit's three main groups and the related PCB's, seen from the left side.

All modules are drawn on one main schematic for easy overview of the complete power supply, see the end of the manual at the drawings section.
Chapter 8 contains a list of all relevant schematics.
Chapter 10 contains the parts list for the components at the main schematic.

6.1 The individual modules

6.1.1 The Control Module

To access
To access the control module/Board:
1. Remove the 4 panel lock screws on the front panel and pull out the MPS.
2. Remove the 2 umbraco screws on top of the front panel.
3. Loosen the 2 umbraco screws on the side panels, and fold down the front panel.
4. Remove the 2 screws from the cover, pull the cover free, and lift it out.
5. Disconnect the six D-type connectors; For inspection or fuse replacement only, just take the cables aside.

This allows for a very quick replacement of all control electronics, thus keeping down the service-time and total downtime.

The control module has the following functions:
- Aux. power for all control circuits.
- Main CPU control of all commands (Remote and Local).
- Display of status.
- Local operation push-buttons.
- Analog current regulation.
- Reference/DAC with temperature control.
- Measuring of output, input and internal voltages/currents.
- Interlock control.

Fig. 6.1.1-1 shows the physical layout of the Control Board PCB:

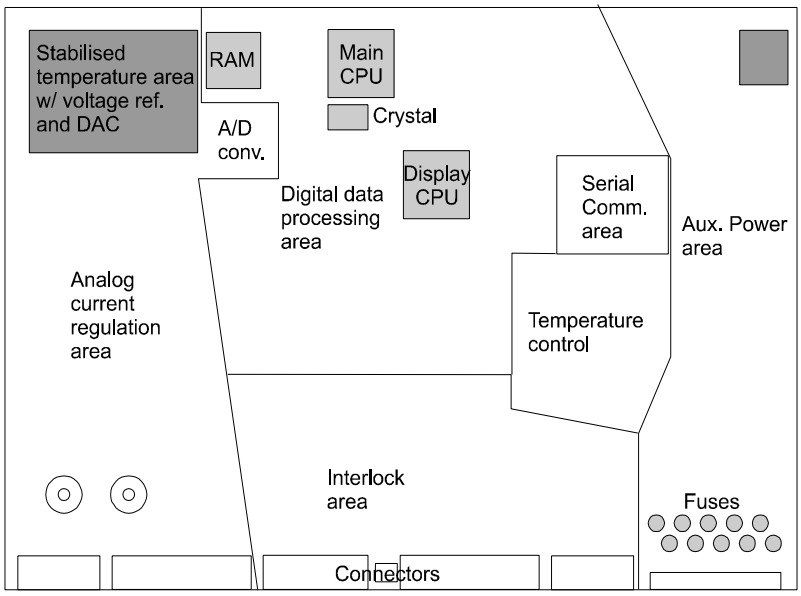


Fig 6.1.1-1 The Control Board

On Fig. 6.1.1-2, a block schematic containing the main functions on the Control Board is shown.

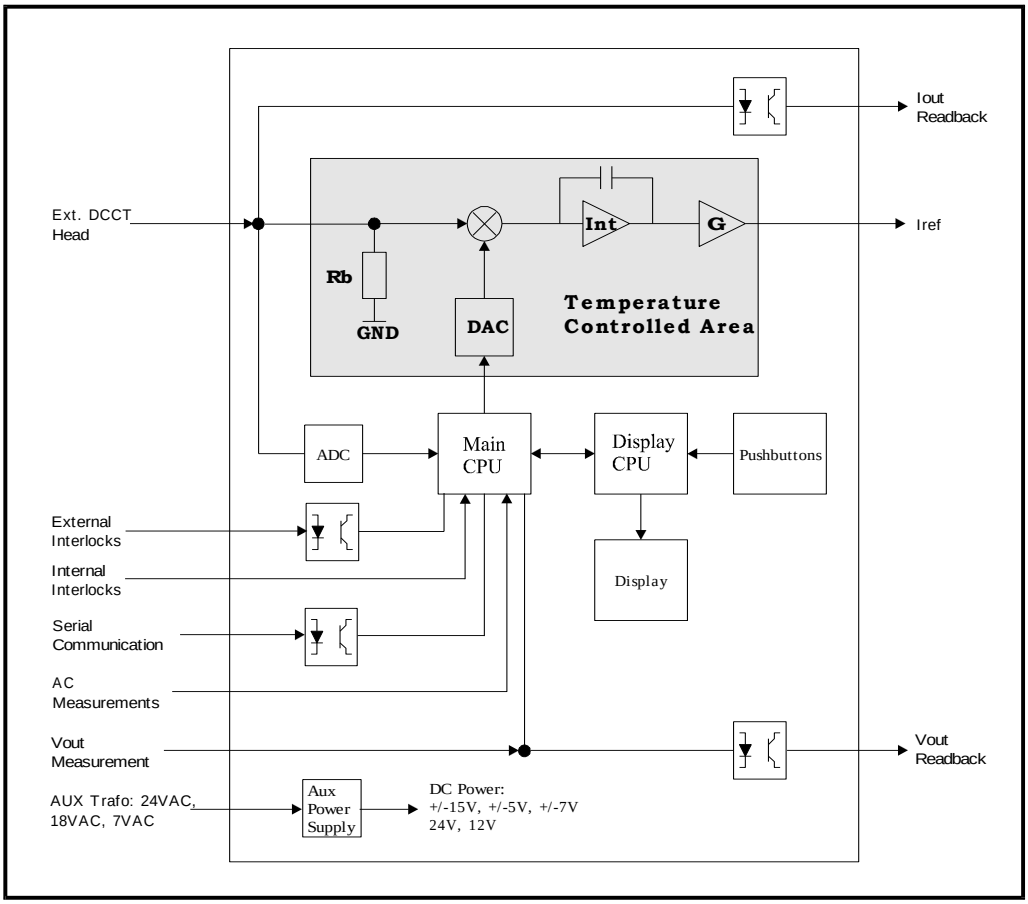


Fig. 6.1.1-2 Block schematic of the Control Board

Aux. power supply: The transformer AUX Trafo for all control board aux. power is located on the AC module and fed to the control module as 24V, 18V and 7V AC power. The rectification and regulation for all supplies is placed on the control board.

Main CPU: An SGS ST9 CPU is used as the main CPU. This CPU has a 16 kB build-in EPROM and four 8 bit I/O ports and eight 8 bit ADCs, see "Measurements", below. The main CPU controls all commands for the supply (remote and local) and all interlocks.

The display is controlled by another ST9 CPU. It controls all readouts to the display and scans the pushbuttons on the frontpanel. The two CPUs are communicating via a local bus.

Measurements: The main CPU's eight 8-bit ADCs are used to measure the AC inputs (current and voltage), output voltage, etc. An 18-bit (Maxim 132) ADC is used for the high precision measurement of the output current, getting its reference from the temperature controlled main reference.

The voltage reference, DAC, Burden resistor and the error amplifier are all placed in a temperature controlled box where the temperature is kept stable within +/- 0.2°C. The controlling element is a Peltier element which can cool as well as heat.

When a new current set value is set, the supply will ramp to this new value with the selected ramp speed after a trig. command has been received. In local mode the trig command is the start push button on the front, in remote control a software trig command is used. Further more, in local as well as remote mode, the MPS can be trigged from the external TTL hardware signal connector (BNC connector on the rear of the supply, completely isolated from the supply).

The analog current regulation is performed by various high precision amplifiers in order to make a high open loop gain ($>10^6$) feedback system. Further details is shown on the schematics for the control board, analog regulation section.

Interlocks are divided into two, external interlocks from outside of the power supply and internal interlocks. The external interlocks are all opto isolated (2.5kV) from the power supply. The voltages remote data communication are insulated (2.5 kV) from all other voltages too.

6.1.2 The Switch Mode power Converter (SMC) module

To access
The Switch Mode power Converter module is located in the centre module of the supply. Like the control module it can be removed for easy replacement:

1. Remove the 4 panel lock screws on the front panel and pull out the MPS.
2. Fold the front panel down, see the description in section 6.1.1, point 2 to 3.
3. Remove the four umbraco screws on the cover plate covering the SMPC module at the centre of the MPS unit. The module is now accessible, see Fig. 6.1.2-1 below.
4. For service or replacement, disconnect the module from the the input/ output cables and the cooling water connection, and lift it out. The module can then be serviced and tested on the bench.

The Swich Mode power Converter module has the following functions:
- Power rectifier bridge for the three AC lines.
- DC link power filter.
- IGBT Full bridge and IGBT driver boards.
- Power transformer
- Secondary power rectifier and output filter board.
- Switch Mode Power Converter (SMPC) Regulation board
- Primary current LEM transducer board

Fig. 6.1.2-1 shows a block diagram of the SMC module; A more detailed overview is found in the MPS main schematic and sub-schematics located at the end of the manual:

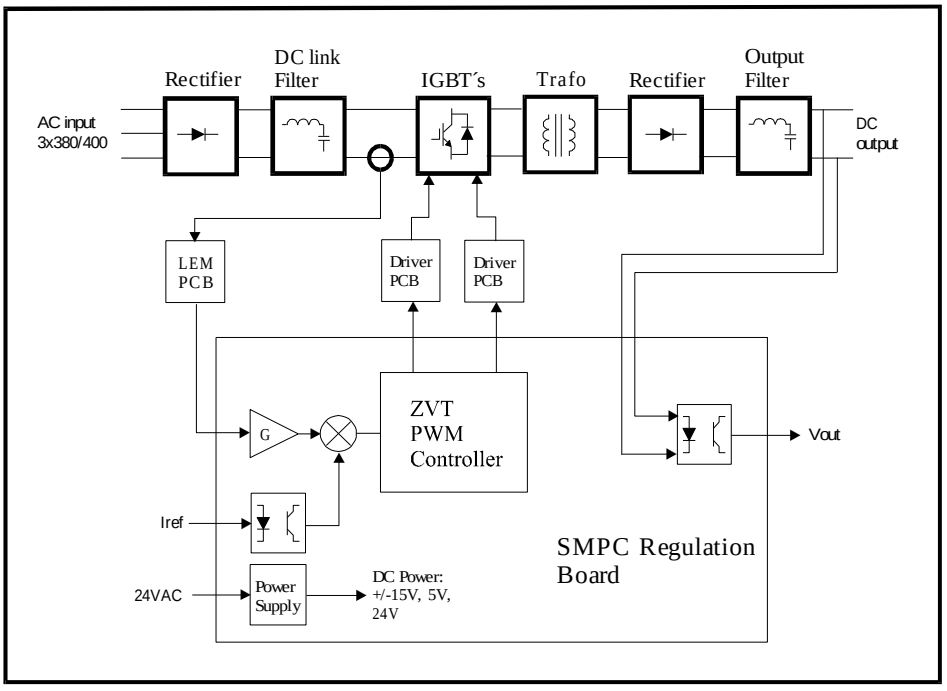


Fig. 6.1.2-1 Block schematic of Switch Mode power Conversion module

Power AC-line rectifier: The AC line rectifier is a 3-phase full-bridge diode rectifier, rated for 60A. Before the rectifier (not shown) is placed an AC line common mode filter to improve the power factor of the diode bridge rectifier and reduce line distortion.

DC link filter: The DC link filter consist of a water-cooled choke input filter and a bulk electrolyte capacitor having the dual function of filtering 300 Hz ripple as well as high frequency noise.

IGBT Full-Bridge: The power switching is performed by two water-cooled, individually controlled IGBT-modules connected as a full-bridge converter.
The main noise sources are the IGBTs and the rectifying diodes in the secondary circuit. A very high efficiency is normally obtained by using fast risetimes. However, this also gives the highest noise level.
Therefore the outstanding performance of the phase shifted, ZVT full-bridge converter topology is chosen. Fully resonant soft-switching waveforms are obtained, combining the advantages of Zero Voltage Transition (ZVT) to reduce switching losses / switching noise and achieving trapezoidal current waveforms in the transistors, transformer and secondary rectifier diodes thus avoiding hard snap off's in the power converting components.
An inductor (not shown), known as the IGBT choke, is placed between the IGBT full bridge and the primary transformer winding. This is done to assure soft-switching even at partial load conditions.
Power transformer: The transformer is a water-cooled, multicoil copper foil transformer wound on a ferrite core. The transformer has four, parallel connected secondary sections, each with a double rectifier. The primaries of each section are series connected for a near ideal current sharing among the parallel connected rectifiers. The secondary windings are shielded for further reduce of the noise transmission between the primary and secondary section. The transformer is made with a certain stray inductance which will aid the switching and give a low winding to winding capacitance resulting in reduced common mode noise currents.

Secondary rectifier: The secondary diode rectifier consist of four dual diode rectifier modules, one for each secondary winding of the transformer. The secondary rectifier diodes may also be used as freewheeling diodes. All rectifier diodes will be in parallel when the magnet is freewheeling, thus giving a very high margin for failure. All eight diodes are rated for at least 100 A each, giving a total of 800 A freewheeling capacity.

Output filter: The output filter choke is split up in two sections, one for each two secondary circuits. The output choke is water cooled and copper foil wound to decrease copper losses and proximity effects. The output filter capacitor bank consists of one hundred ceramic capacitors in parallel, thus decreasing the un-wanted parasitic components, and thereby improving the overall efficiency and filtering performance.

LEM PCB: The LEM PCB is the primary current monitor unit used for primary current regulation and over current detection.

IGBT Driver PCB: Two IGBT driver boards are used, one for each IGBT module. Each IGBT driver board provides isolated low side and high side drivers (one of each) capable of driving up to 2A. Further more isolated signals indicating the status of the IGBT's and the IGBT driver are given. These status signals is seen as two green LED's on each board, these should always be lit indicating that IGBT's and drivers are OK.

Switch Mode Power Converter (SMPC) Regulation board:
The SMPC Regulation board is a very vital part of the switch mode power supply. This board provides switching schemes for the power conversion and takes care of the overcurrent protection and shut-down procedures.
The SMPC Board is provided with its own power supply to increase noise immunity and to maintain isolation demands.
The heart of the SMPC Regulation board is the Phase modulated PWM controller ML4818. This IC provides a fully phase modulation technique for full-bridge soft-switching transitions, programmable delay-time, pulse-by-pulse current limit and soft-start procedure. Further more undervoltage lock-out and shut-down facilities are available and used.
The PWM controller is operated in current control mode, and the switching is performed as follows, referring to Fig. 6.1.2-2 and Fig. 6.1.2-3:

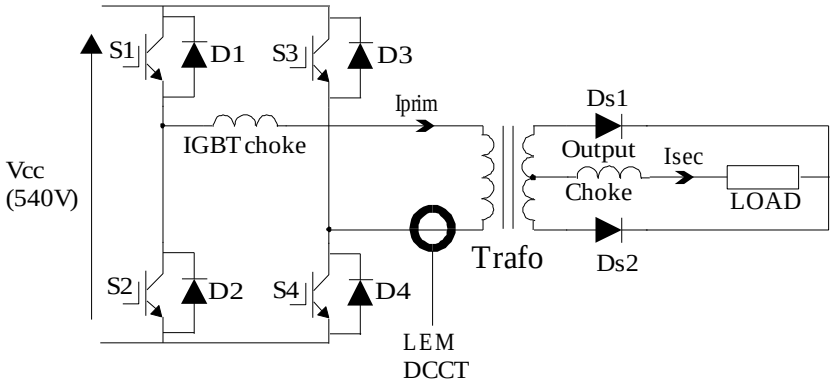


Fig. 6.1.2-2 Principle diagram of the switch mode power converter

The switches S1 and S2 are running at 20kHz clock in opposite phase, ie. the switching period Tsw is 50 usec. To avoid shoot-through in the IGBT's, there is a small time delay between the turn off and turn on of the switches. S3 and S4 are running in opposite phase too with a small transition delay. The primary current is controlled by the delay between the switch pairs S1/S2 and S3/S4. The switching of S3/S4 is controlled by a comparison between the desired set current value and the primary current level measured by the LEM DCCT in the the primary current loop.
The secondary side consists of a centre-tap winding where the output choke is connected, and two output rectifier diodes Ds1 and Ds2 are connected to a full-bridge rectifier. Physically, there is four secondary winding and diode sections in parallel.

Fig. 6.1.2-3 shows the switching principle and the belonging switching waveforms:

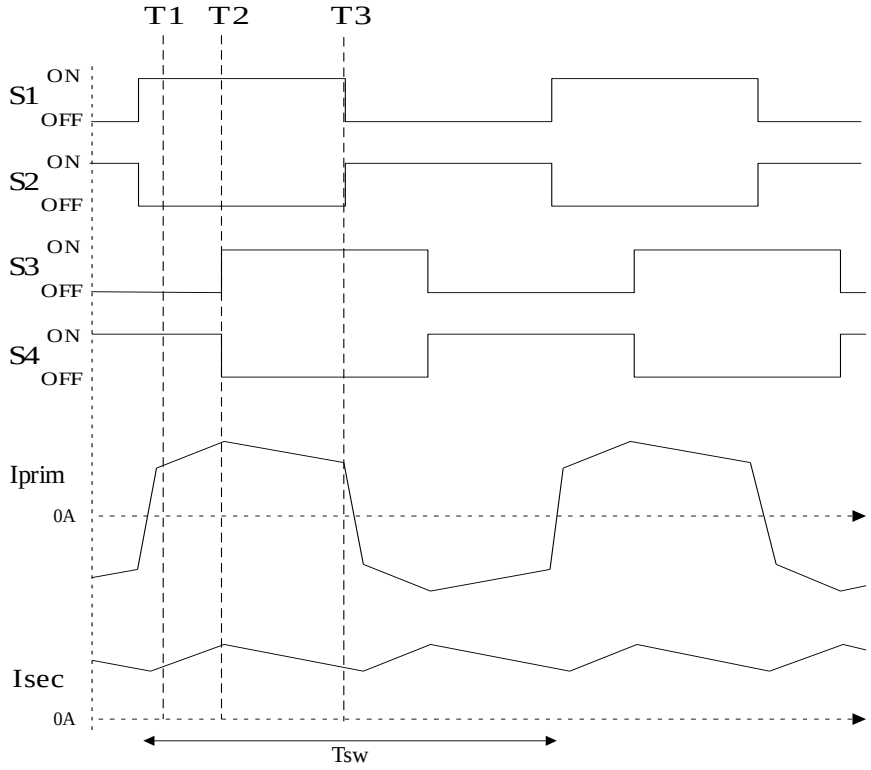


Fig. 6.1.2-3 Switching waveforms of the switch mode power converter

Time T1: Consider that the IGBT´s S1 and S4 are "on", ie. Vcc is impressed the primary winding. Primary current is increasing, and energy is transformed from the primary to the secondary. The secondary current is flowing through the load and Ds1and is increasing.
When the primary current measured by the LEM DCCT reaches the desired set current value, S4 is turned off.

Time T2: S4 is switched off, now the primary current is running in D3 and S1. The primary current is free-wheeling by the energy stored in the IGBT choke, and energy is still transferred to the secondary side, but the primary current as well as the secondary current is decreasing.
When S4 is turned off and after a short delay, S3 is turned on, but due to the fact that the current is free-wheeling in D3, S3 is turned on but there is no voltage across S3! Zero Voltage Switching (ZVT) is obtained!

Time T3: IGBT S1 is turned off by the 20kHz timer. The primary current is now running through D2 and S3, and the current is decreasing very rapidly due to the 540V it is free-wheeling “against”. No energy is transferred to the secondary, and the secondary current is free-wheeling through both diodes due to the energy stored in the output choke.
After a short delay of the turn off of S1, S2 is turned on, but D2 is still conducting, ie. There is no voltage across S2, and Zero Voltage Switching (ZVT) again is obtained!
Now the current commutates from freewheeling through D2 to conducting throug S2 and S3, and a new switch cycle is started.

The improvements achieved of Zero Voltage Switching versus the classical hard switching system are:

- Trapezoidal current waveforms in the IGBT's, transformer and secondary rectifier diodes avoiding hard snap-off, and thereby decreasing the noise level
- Primary transformer winding is always clamped avoiding primary side oscillations
- Turn-on losses of the IGBT's are eliminated, giving improved efficiency

The physical lay-out of the SMPC regulation board is shown on Fig. 6.1.2-4

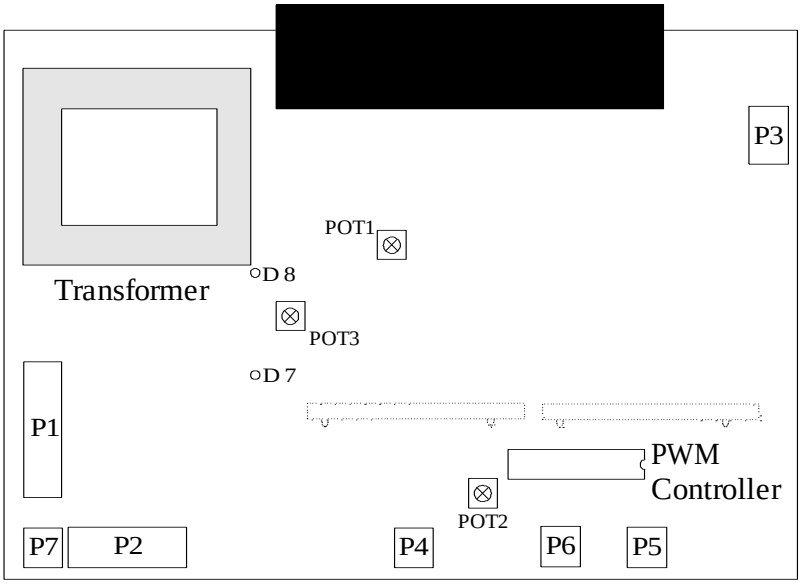


Fig. 6.1.2-4. The SMPC Regulation Board

At the SMPC Regulation Board, two green LED's are mounted, D7 and D8. D7 indicates, when lit, that the onboard 5V power supply is OK. D8 indicates, when lit, that the onboard +/- 15 power supply is present, and the connection to the LEM DCCT board is OK.

Note that when mounted in the power supply, below the SMPC Board, the two small IGBT Driver Boards (dashed lines) are placed vertically. Each of these carries two green LEDs. These should always be lit, indicating that the IGBT's are OK.

6.1.3 AC mains input module

As for the other modules, the AC mains input module is located in a box which is easy to remove for service or replacement.

To access

1. Open the back door of the cabinet by turning the 3 screws on the back door 90̊ counterclockwise.
2. Remove the AC mains input module cover: Turn the two black plastic screws on top of the panel 90̊ in any direction and remove the cover. The AC Interface Board is placed just behind the connectors at the top of Fig. 6.1.3-1 with its component side down.

The AC module has the following functions:

- Connections for incomming AC lines.
- Adjustable max. current breaker with undervoltage relay.
- Main contactor with soft-start relay.
- AC current and AC voltage measurements.
- Control power contactor.
- AC Interface Board handling output current / output voltage readback, interlock connectors and serial communication connectors.

All the mains power components are located in this module.

Regarding the AC power flow in the AC module, see the AC Module wiring diagram located at the drawings section at the end of the manual.

Starting from the incomming mains the flow is:

a) 3 small-power transformers, used for voltage measurement and feeding the “mains present” LEDs on the front panel. These transformers are located on the AC Interface Board and fused with 0.1 A fuses with a breaking capacity of up to 30 kA. Depending on the time of manufacturing of the power supply, the signal for the 3 measuring transformers might be taken after the main circuit breaker, please consult the AC Module wiring diagram found at the end of the manual.
b) The adjustable max. Current breaker with undervoltage relay for remote release. The circuit breaker trip level is factory set and must not be changed. Interrupting capacity of the breaker is up to 30 kA, depending on model type.
c) 3 small current transformers, used for mains current monitoring.
d) The main contactor with a soft start contactor. The soft start contactor connects three 3 ohm resistors in series with the mains for approx. 15 msec. to limit the inrush current.

The Aux.- power transformer providing low power AC voltage power supply for the electronics are located at the bottom of the AC module. The fused aux.-power transformer has a manual circuit breaker for service disconnection.

As earlier described the AC Interface board is handling the routing of readback signals from the power supply to the external connectors on the rear as well as control signals for the power supply going in the opposite direction. Further more all control signals for the AC power relays go via the AC interface board, where all signals are routed to their appropriate destinations. The diagram of the

AC interface Board is shown in at the end of the manual.

Fig. 6.1.3-1 shows the physical lay-out of the AC module:

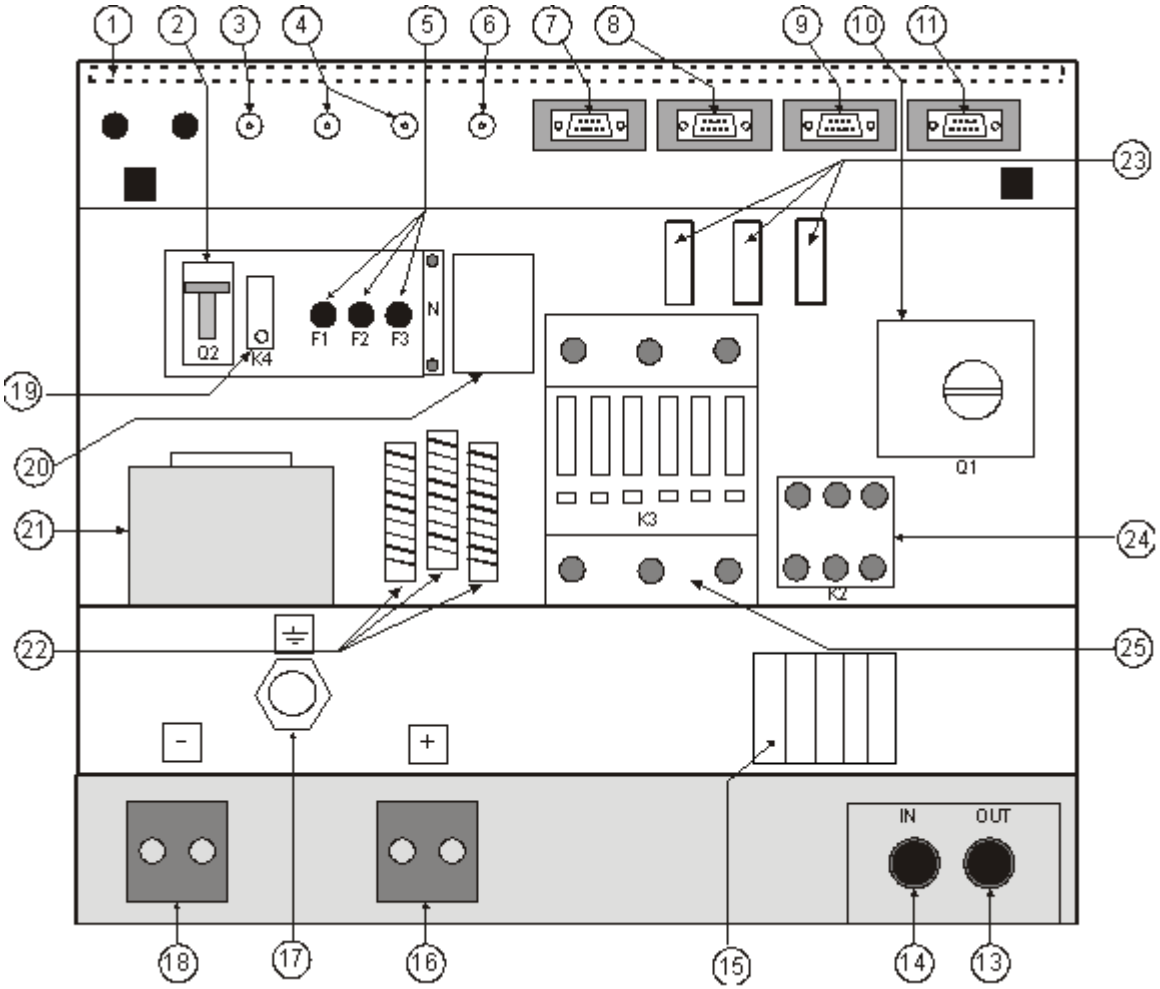


Fig. 6.1.3-1 The AC mains input module with cover removed.

1. AC Interface Board
2. Control voltage breaker Q2. Shown in ON position
3. Voltmeter: output voltage: 0V=0%. 10V=100% of max voltage.
4. Voltmeter: output current: 0V=0%. 10V=100% of max current.
5. Fuses for AC voltmeter and front panel R, S, and T phase LEDs.
6. Trigger input signal. TTL levels. 0->1 = Start.
7. Rack interlocks. Optically insulated 0-24V levels. See further details in section 2.4
8. User interlocks. Optically insulated 0-24V levels. See further details in section 2.4
9. RS232C connector. Pin-for-pin connected to (11). See section 2.4.4.
10. Main on/off switch.
11. RS232C connector. Pin-for-pin connected to (13). See section 2.4.4.
12. NA
13. Cooling water out.
14. Cooling water in.
15. Power input 1. When mounted in rack as shown. Concerning single MPS installations the connector is a terminal strip marked R, S, T, Neutral and Earth.
16. Magnet current output terminal (+).

17. Ground connection
18. Magnet current output terminal (-).

19. Power on rely.
20. Flow switch converter.
21. Aux.-transformer.
22. Inrush-current limiting resistors.
23. AC current measurement transformers.
24 Soft-start relay.
25. Main contactor power relay.

6.1.4 Function description of Printed Circuit Board's.

Control Board (Section 6.1.1) has following functions:

- Aux. power for the control module circuits.
- Main CPU control of all commands (Remote and Local).
- Display of status (local).
- Local operation push-buttons
- Analog current regulation.
- Reference/DAC with temperature control.
- Measuring of output, input and internal signals.
- Interlock control.

SMPC Regulation Board (Section 6.1.2) has following function:

- PWM (Pulse Width Modulator) controller.
- Current regulator (Primary switch current).
- Make isolation between control and power components.

LEM DCCT Board (Section 6.1.2) has following function:

- Measurement of primary IGBT / transformer current.

Output filter Board (Section 6.1.2) has following function:

- Carrying capacitors for output filter.
- EMI earthing.
- Connection for output voltage measurement.

AC Interface board (Section 6.1.3) has following functions:

- Measurement of line input voltage.
- Measurement of line input current.
- Measurement of ground leak current.
- Output current and output voltage read-back connectors.
- Serial communication connectors
- External interlock connectors
- Connections between in and out side of the power supply

6.2 The Interlock system

An "interlock" is a condition which has stopped the power supply, and which prevents it from being started again before the cause of the interlock has been removed.

All interlocks are shown on the front panel display and can be read from the remote control terminal. If there are more interlocks than can be shown on the front panel display at once, the up and down pushbuttons can be used to scroll up and down through the list.

There are five groups of interlocks:
- Status interlocks. Manually set conditions preventing the MPS from starting.
- AC input module interlock.
- Switch mode Power Converter interlocks.
- Control module interlocks.
- User interlocks. User definable interlocks, e.g. connected to a magnet thermal breaker or a magnet cooling water flow switch.

Status interlocks:
- Door open
- Neutral polarity; Shown in the DC menu as NEU
- Low water flow

AC input module interlock:
- AC Undervoltage
- Ground leak current (warning)
- Freewheel diode

Switch Mode power Converter interlocks:
- IGBT heatsink overtemp.
- H.F. diode heatsink overtemp.
- Switch regulator supply failure
- Switch regulator DCCT failure
- IGBT driver failure
- Chassis overtemp.
- Over-current

User interlocks:
- 6 User definable interlocks, opto isolated and available on a D-sub connector on the rear of the supply, see section 2.4.3.

6.3 Connectors

6.3.1 Control board

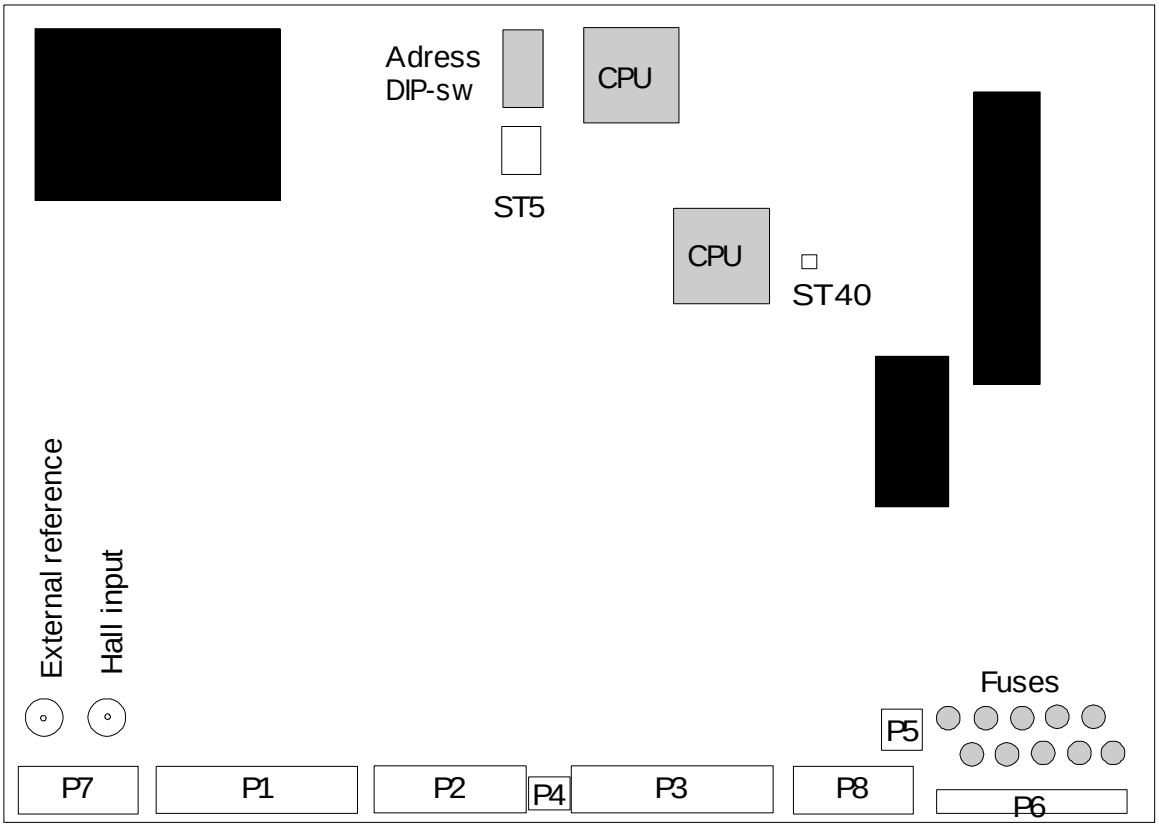


Fig. 6.4.1-1 The control board connectors

The Control board has the following connectors:

-P1 SMPC Regulation board interface
-P2 AC Interface board interface
-P3 AC Interface board interface
-P4 Emergency switch interface
-P5 Optional service key interface
-P6 Aux - power supply
-P7 DCCT interface
-P8 AC Interface board interface

Connector P1: (To/From SMPC Regulation board connector P1)

Pin:	Name	I/O	Description & Specification
1	ICTL/P11	O	Voltage high side for high precision current regulation.
2	ICTL/P12	O	Voltage low side for high precision current regulation.
3	VOUT/AN0	I	Voltage representing the voltage at the output terminals.
4	NC		Not used.
5	NC		Not used.
6	AC1L	O	24VAC Power supply.
7	NC		Not used.
8	NC		Not used.
9	INT38	O	Power Is On
10	INT22	I	IGBT driver failure interlock
11	INT20	I	Switch reg. DCCT failure interlock
12	INT18	I	IGBT heatsink overtemp. interlock.
13	INT16	I	Diode heatsink overtemp. Interlock
14	A+15V	O	+15V Power supply.
15	GNDA15V	O	Return 15V Power supply.
16	A-15V	O	-15V Power supply
17	NC		Not used.
18	AC1N	O	24VAC Power supply.
19	NC		Not used.
20	NC		Not used.
21	GND	O	Isolated ground for interlocks signals.
22	INT23	I	Spare interlock
23	INT21	I	Switch regulator supply failure interlock
24	INT19	I	H.F. diode overtemp. interlock
25	INT17	I	Chassis overtemp. interlock

Connector P2: (To/From AC Interface board connector P4)

Pin:	Name	I/O	Description & Specification
1	G_ LEAK/AN1	I	Voltage represent current that is leaking to ground
2	NC	Not used.	
3	A-15V	O	-15V Power supply.
4	NC	Not used.	
5	ACRTN/GNDA15V	O	Return 15V power supply and AC measurements
6	IAC2/AC4	I	Voltage represent input current phase S.
7	VAC3/AC2	I	Current represent input voltage phase T.
8	VAC1/AC0	I	Current represent input voltage phase R.
9	GNDA15V	O	Return 15V Power supply.
10	GNDA15V	O	Return 15V Power supply.
11	A+15V	O	+15V Power supply.
12	GNDA15V	O	Return 15V Power supply.
13	IAC3/AC5	I	Voltage represent input current phase T
14	IAC1/AC3	I	Voltage represent input current phase R
15	VAC2/AC1	I	Current represent input voltage phase S

Connector P3: (To/From AC Interface board connector P8)

Pin:	Name	I/O	Description & Specification
1	+24V	O	+24V Power supply external interlocks.
2	FREE1/INT12	I	Free 1 interlock
3	POL+/ INT10	I	Positive polarity
4	THERM/INT8	I	Low waterflow interlock
5	USER6/INT6	I	User 6 interlock
6	USER4/INT4	I	User 4 interlock
7	USER2/INT2	I	User 2 interlock
8	ISON/INT33	I	Power is on signal +24V when main contactor is ON.
9	GOFF	O	Turn OFF the max. current breaker.
10	TRIG-/INT34	I	Low side trigger for start ramping (power supply is ON)
11	C-15V	O	-15V C Power supply.
12	C GND/GNDC15V	O	Return C Power supply.
13	VOUT	O	Voltage 0-10V representing output voltage 0-100%
14	S.INTL/INT37	O	Sum interlock low when an interlock is present.
15	POL-/INT11	I	Negative polarity
16	DOOR/INT9	I	Door interlock
17	PHASE/INT7	I	Free-wheel diode thermal interlock.
18	USER5/INT5	I	User 5 interlock.
19	USER3/INT3	I	User 3 interlock
20	USER1/INT1	I	User 1 interlock
21	MAINON/INT32	O	Indicate that main is ON.
22	TRIG+/INT35	I	High side trigger for start ramping (power supply is ON)
23	GND24V	O	Return 24V Power supply.
24	C+15V	O	+15V C Power supply.
25	IOUT	O	Voltage 0-10V representing output current 0-100%

Connector P4: (Emergency switch interface)

Pin:	Name	I/O	Description & Specification
1	INT36	O	Connection to the emergency switch.
2	GOFF	I	Connection to the emergency switch.

Connector P5: (Service key interface)

Pin:	Name	I/O	Description & Specification
1	C12		Connection to the set-up key switch.
2	GND		

Connector P6: (Aux - power supply)

Pin:	Name	I/O	Description & Specification
1	24VAC/ACIL	I	24VAC.
2	24VAC/ACIN	I	24VAC.
3	NC		Not used.
4	A-PSU18VAC/AC2LH	I	18VAC high side A Power supply
5	A-PSU18VAC/AC2N	I	Return 18VAC A Power supply.
6	A-PSU18VAC/AC2LL	I	18VAC low side A Power supply.
7	7VAC/AC3H	I	7VAC high side Power supply Peltier element.
8	7VAC/AC3N	I	Return Power supply Peltier element.
9	7VAC/AC3L	I	7VAC low side Power supply Peltier element.
10	B-PSU18VAC/AC4LH	I	18VAC high side B Power supply.
11	B-PSU18VAC/AC4N	I	Return B Power supply.
12	B-PSU18VAC/AC4LL	I	18VAC low side B Power supply.
13	NC		Not used.
14	C-PSU18VAC/AC5LH	I	18VAC high side C Power supply.
15	C-PSU18VAC/AC5N	I	Return C Power supply.
16	C-PSU18VAC/AC5LL	I	18VAC high side C Power supply.

Connector P7: (DCCT interface)

Pin:	Name	I/O	Description & Specification
1	NC		Not used.
2	NC		Not used.
3	/P73	I	Signal to DCCT over-current interlock
4	GNDB/P74	I/O	Output current low side DCCT. 0.4A source current when DCCT head is 100% loaded. Return 15V Power supply.
5	B-15V	O	-15V Power supply for DCCT head.
6	/P76	I	Output current high side DCCT. 0.4A source current when DCCT head is 100% loaded.
7	NC		Not used.
8	/P78	I	Over current interlock.
9	B+15V	O	+15V Power supply for DCCT head.

Connector P8: (To/From AC Interface board connector P16)

Pin:	Name	I/O	Description & Specification
1	TXH/RS2	O	Transmit line high side of the remote RS 422 port.
2	TXL/RS3	O	Transmit line low side of the remote RS 422 port.
3	RXH/RS0	I	Receive line high side of the remote RS 422 port.
4	RXL/RS1	I	Receive line low side of the remote RS 422 port.
5	TXCOM/ISOGND	I/O	Isolated ground for the serial interface.
6	NC		Not used.
7	NC		Not used.
8	NC		Not used.
9	TXVCC/ISOVCC	O	Isolated +5V for the serial interface.

6.3.2 SMPC Regulation board

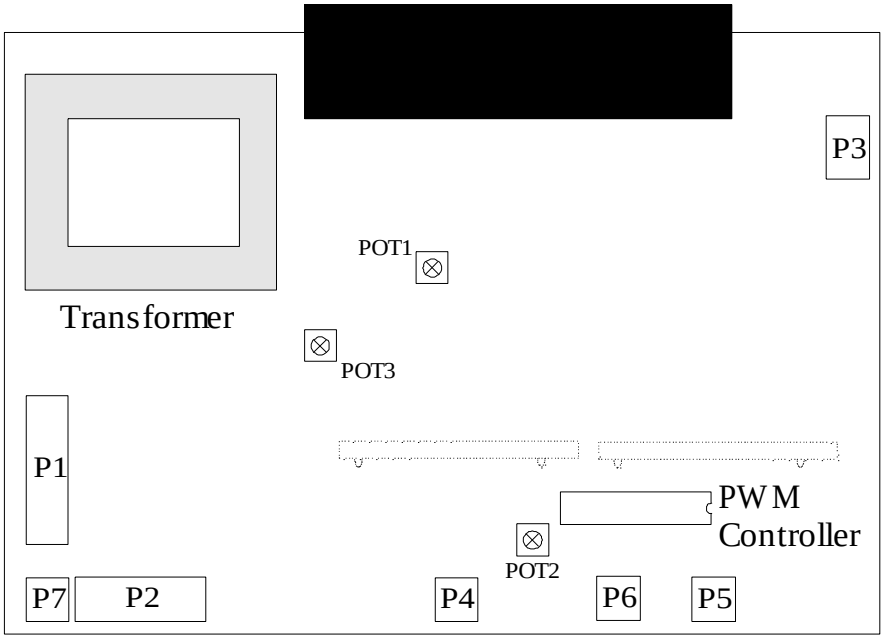


Fig. 6.4.2-1 The SMPC Regulation board

The SMPC Regulation board has the following connectors:

-P1 Control board interface
-P2 Thermal switch interlock interface
-P3 LEM-DCCT board interface
-P4 Output filter board
-P5 IGBT driver module 1 interface
-P6 IGBT driver module 2 interface
-P7 24VDC Blower

Connector P1: (To/From Control board P1)

Pin:	Name	I/O	Description & Specification
1	ICTL/P1-1	I	Voltage high side for high precision current regulator.
2	NC		Not used.
3	VOUT/P1-3	O	Voltage representing the voltage at the output terminals.
4	NC		Not used.
5	NC		Not used.
6	/P1-6	I	24VAC Power supply.
7	NC		Not used.
8	NC		Not used.
9	/P1-9	I	Power is ON.
10	/P1-10	O	IGBT driver failure interlock, low when active.
11	/P1-11	O	Switch reg. DCCT failure interlock, low when active.
12	/P1-12	O	IGBT heatsink overtemp. Interlock, low when active.
13	/P1-13	O	Diode heatsink overtemp. Interlock, low when active.
14	/P1-14	I	+15V Power supply.
15	/P1-15	I	Return 15V Power supply.
16	/P1-16	I	-15V Power supply.
17	NC		Not used.
18	/P1-18	I	24VAC Power supply.
19	NC		Not used.
20	NC		Not used.
21	/P1-21	O	Isolated return for interlocks signals.
22	/P1-22	O	Spare interlock, low when active.
23	/P1-23	O	Switch regulator supply interlock, low when active.
24	/P1-24	O	H.F. Diode overtemp. interlock, low when active.
25	/P1-25	O	Chassis overtemp. Interlock, low when active.

Connector P2: (Thermal switch interlock)

Pin:	Name	I/O	Description & Specification
1	/P2-1	I	High side thermal switch on diode heatsink.
2	24GND	O	Low side thermal switch on diode heatsink.
3	/P2-3	I	High side thermal switch on chassis.
4	24VGND	O	Low side thermal switch on chassis.
5	/P2-5	I	High side thermal switch on IGBT heatsink.
6	24VGND	O	Low side thermal switch on IGBT heatsink.
7	/P2-7	I	High side thermal switch on H.F. diode heatsink.
8	24VGND	O	Low side thermal switch on H.F. diode heatsink.

Connector P3: (To/From LEM-DCCT board connector P1)

Pin:	Name	I/O	Description & Specification
1	+/-15V ok &DCCT ok	I	Signal indicate power supply and LEM DCCT board is connected.
2	A-15	O	+15V Power supply.
3	Current	I	Current signal represent primary current.
4	A+15	O	-15V Power supply.

Connector P4: (To/From Output filter board connector P1)

Pin:	Name	I/O	Description & Specification
1	+Vout/Vout+	I	High side of voltage representing voltage at output terminals.
2	-Vout/vout-	I	Low side of voltage representing voltage at output terminals.

Connector P5: (To/From IGBT driver module 1)

Pin:	Name	I/O	Description & Specification
1	A+15	O	+15V Power supply.
2	AGND	O	Return 15V Power supply.
3	DR-fail A	I	Driver failure on driver A.
4	Gate A	O	Gate signal driver A
5	DR-fail B	O	Gate signal driver B
6	Gate B	I	Driver failure on driver B
7	B+5	O	+5V Power supply.
8	BGND	O	Return 5V Power supply.

Connector P6: (To/From IGBT driver module 2)

Pin:	Name	I/O	Description & Specification
1	A+15	O	+15V Power supply.
2	AGND	O	Return 15V Power supply
3	DR-fail A	I	Driver failure on driver A.
4	Gate A	O	Gate signal driver A
5	DR-fail B	O	Gate signal driver B
6	Gate B	I	Driver failure on driver B
7	B+5	O	+5V Power supply.
8	BGND	O	Return 5V Power supply

Connector P7: (To 24V blower)

Pin:	Name	I/O	Description & Specification
1	+24VDC	O	+24V Power supply.
2	-24VDC	O	Return 24V Power supply.

6.3.3 AC Interface board

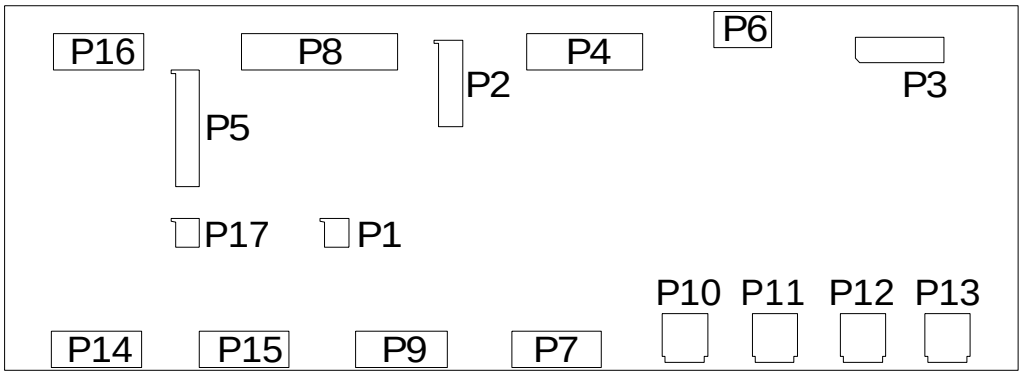


Fig. 6.4.3-1 The AC interface board.

The AC Interface board has the following connectors:

-P1 Ground leak measurement
-P2 AC current transformer interface
-P3 AC line voltage input plug interface
-P4 Control board interface
-P5 Main contactor interface
-P6 Interlocks switches interface
-P7 External interlocks interface
-P8 Control board interface
-P9 User interlocks interface
-P10 Ramp start interface
-P11 Iout (0-10V) interface
-P12 Iout (0-10V) interface
-P13 Vout (0-10V) interface
-P14 Remote line serial communication (Default RS422)
-P15 Remote line serial communication (Default RS422)
-P16 Control board interface (Default RS422)
-P17 Free interlock interface

Connector P1: (Ground leak measurement)

Pin:	Name	I/O	Description & Specification
1	GND	I	CHASSIS
2	Measurement	I	Measurement of current leak to ground.

Connector P2: (To/From AC current transformer)

Pin:	Name	I/O	Description & Specification
1	RTN_ac	I/O	Return current-transformer phase T.
2	T_I ac	I	High side current-transformer phase T.
3	RTN_ac	I/O	Return current-transformer phase S.
4	S_I ac	I	High side current-transformer phase S.
5	RTN_ac	I/O	Return current-transformer phase R.
6	R_I ac	I	High side current-transformer phase R.

Connector P3: (To/From AC line voltage)

Pin:	Name	I/O	Description & Specification
1	RTN	I/O	Return voltage-transformers.
2	TFASE	I	High side voltage-transformer phase T.
3	SFASE	I	High side voltage-transformer phase S.
4	RFASE	I	High side voltage-transformer phase R.

Connector P4: (To/From Control board connector P2)

Pin:	Name	I/O	Description & Specification
1	GND LEAK/G Leak	O	Voltage represent current that is leaking to ground.
2	NC		Not used.
3	A-15V	I	-15V Power supply.
4	NC		Not used.
5	ACRTN/GNDA15V	O	Return 15V Power supply and AC measurements.
6	IAC2/S I ac	O	High side current-transformer phase S.
7	VAC3/T V ac	O	High side voltage-transformer phase T.
8	VAC1/R V ac	O	High side voltage-transformer phase R.
9	GNDA15V	I/O	Return DC and 15V Power supply.
10	GNDA15V	O	Return 15V Power supply.
11	A+15V	I	+15V Power supply.
12	GNDA15V	I	Return 15V power supply.
13	IAC3/T I ac	O	High side current-transformer phase T.
14	IAC1/R I ac	O	High side current-transformer phase R.
15	VAC2/S Vac	O	High side voltage-transformer phase S.

Connector P5: (To/From Main contactor)

Pin:	Name	I/O	Description & Specification
1	Main ON	O	Low side signal to turn main ON.
2	+24V	O	High side signal to turn main ON.
3	Main is ON	I	High side signal indicate that main is ON.
4	+24VRTN	O	Low side signal indicate that main is ON.
5	Circuit br.	O	Low side signal to hold max. Current breaker ON.
6	+24V	O	High side signal to hold max. Current breaker ON
7	NC		Not used.
8	+24V	O	Not used.

Connector P6: (To Interlocks switches)

Pin:	Name	I/O	Description & Specification
1	Flow switch/Phase	I	High side thermal switch free-wheel diode overtemp. interlock.
2	+24VRTN	O	Low side thermal switch free wheel diode overtemp. interlock.
3	Free wheel/Thermal	I	High side waterflow interlock.
4	+24VRTN	O	Low side waterflow interlock

Connector P7: (To/From External interlocks)

Pin:	Name	I/O	Description & Specification
1	DOOR/Door	I	High side door interlock.
2	NORMAL/Normal	I	High side polarity indication positive.
3	REVERSED/Reversed	I	High side polarity indication negative.
4	INT.LIGHT/Int:light	O	Low side sum interlock for indicator on top of rack.
5	NC		Not used.
6	+24VRTN	O	Low side door interlock.
7	+24VRTN	O	Low side polarity indication positive.
8	+24VRTN	O	Low side polarity indication negative
9	+24V	O	High side sum interlock for indicator on top of rack

Connector P8: (To/From Control board connector P3)

Pin:	Name	I/O	Description & Specification
1	+24V	I	+24V Power supply for external interlocks.
2	FREE1/Freel	O	Free1 interlock, low when active.
3	POL+/Normal	O	Positive polarity indication. Low when active
4	THERM/Thermal	O	Low waterflow interlock, low when active.
5	USER6/User6	O	User 6 interlock, low when active.
6	USER4/User4	O	User 4 interlock, low when active.
7	USER2/User2	O	User 2 interlock, low when active.
8	ISON/Main is ON	O	Power is ON signal +24V when main contactor is ON.
9	GOFF/Circuit br.	I	Turn OFF the max. Current breaker.
10	TRIG-/TrigRTN	O	Low side trigger for start ramping (power supply is ON)
11	C15VSUPPL/C-15V	I	-15V Power supply.
12	GNDC15V/C GND	I	Return 15V Power supply.

13	VOUT/V(out)	O	Voltage 0-10V representing output voltage 0-100%.
14	S.INT/Suminterlock	I	Sum interlock; low when an interlock is present.
15	POL+/Reversed	O	Negative indication. Low when active.
16	DOOR/Door	O	Door interlock, low when active.
17	PHASE/Phase	O	Free-wheel diode temperature interlock. Low when active.
18	USER5/User5	O	User 5 interlock, low when active.
19	USER3/User3	O	User 3 interlock, low when active.
20	USER1/User1	O	User 1 interlock, low when active.
21	MAINON/Main ON	I	Indicate that main is ON.
22	TRIG+/Trig	O	High side trigger for start ramping (power supply is ON)
23	+24VRTN	I	Return 24V Power supply.
24	C15VSUPPL/C+15V	I	+15V Power supply.
25	IOUT/I(out)	O	Voltage 0-10V representing output current 0-100%.

Connector P9: (To/From User interlocks)

Pin:	Name	I/O	Description & Specification
1	User1	I	High side user1 interlock.
2	User2	I	High side user2 interlock.
3	User3	I	High side user3 interlock.
4	User4	I	High side user4 interlock.
5	User5	I	High side user5 interlock.
6	User6	I	High side user6 interlock.
7	NC		Not used.
8	NC		Not used.
9	+24VRTN	O	+24VDC return for user interlocks.

Connector P10: (Trig signal for ramp start)

Pin:	Name	I/O	Description & Specification
1	Trig	I	TTL positive going signal to start ramping when power is ON.
2	TrigRTN	I	Return trig signal

Connector P11: (Iout 0-10V)

Pin:	Name	I/O	Description & Specification
1	I(out)	O	Voltage 0-10V representing output current 0-100%.
2	C_GND	I/O	Return Iout

Connector P12: (Iout 0-10V)

Pin:	Name	I/O	Description & Specification
1	I(out)	O	Voltage 0-10V representing output current 0-100%.
2	C_GND	I/O	Return Iout

Connector P13: (Vout 0-10V)

Pin:	Name	I/O	Description & Specification
1	V(out)	O	Voltage 0-10V representing output voltage 0-100%.
2	C_GND	I/O	Return Vout

Connector P14: (To Host Remote line, default RS422 shown; Connected in parallel with P15)

Pin:	Name	I/O	Description & Specification
1	TXH/RS8	O	Transmit line high side of the remote RS 422 port.
2	TXL/RS6	O	Transmit line low side of the remote RS 422 port.
3	RXH/RS4	I	Receive line high side of the remote RS 422 port.
4	RXL/RS2	I	Receive line low side of the remote RS 422 port.
5	RETURN/RS0	I/O	Isolated ground for the serial interface.
6	NC		Not used.
7	NC		Not used.
8	NC		Not used.
9	+5V	O	Isolated +5V for the serial interface.

Connector P15: (To Host Remote line, default RS422 shown; Connected in parallel with P14)

Pin:	Name	I/O	Description & Specification
1	TXH/RS8	O	Transmit line high side of the remote RS 422 port.
2	TXL/RS6	O	Transmit line low side of the remote RS 422 port.
3	RXH/RS4	I	Receive line high side of the remote RS 422 port.
4	RXL/RS2	I	Receive line low side of the remote RS 422 port.
5	RETURN/RS0	I/O	Isolated ground for the serial interface.
6	NC		Not used.
7	NC		Not used.
8	NC		Not used.
9	+5V	O	Isolated +5V for the serial interface.

Connector P16: (default RS422 to Control board P8 shown)

Pin:	Name	I/O	Description & Specification
1	TXH/RS8	O	Transmit line high side of the remote RS 422 port.
2	TXL/RS6	O	Transmit line low side of the remote RS 422 port.
3	RXH/RS4	I	Receive line high side of the remote RS 422 port.
4	RXL/RS2	I	Receive line low side of the remote RS 422 port.
5	RETURN/RS0	I/O	Isolated ground for the serial interface.
6	NC		Not used.
7	NC		Not used.
8	NC		Not used.
9	+5V	O	Isolated +5V for the serial interface.

Connector P17: (To Free Interlock)

Pin:	Name	I/O	Description & Specification
1	Free1	O	Not used.
2	+24VRTN	O	Not used.

6.3.4 Output filter board



Fig. 6.4.4-1 The output filter board.

The Output filter board has the following connectors:

-P1 Output voltage measurement interface
-P2 DCCT filter compensation winding
-L4 Snubber interface for output choke
- CHASSIS

Connector P1: (Output voltage measurement to SMPC regulation board P4)

Pin:	Name	I/O	Description & Specification
1	+Vout/vout+	O	High side of voltage output terminals.
2	-Vout/vout-	O	Low side of voltage at output terminals.

Connector P2: (From DCCT compensation winding on DCCT head)

Pin:	Name	I/O	Description & Specification
1	/pin1	I	DCCT compensatiuon winding.
2	/pin2	I	DCCT compensation winding.

Connector L4.x: (From each half of output choke)

Pin:	Name	I/O	Description & Specification
1	L4.1	I	Output choke part 1.
2	L4.2	I	Output choke part 2.

6.3.5 LEM DCCT board

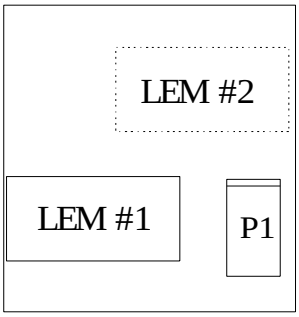


Fig. 6.4.5-1 The LEM DCCT board.

The LEM DCCT board has the following connector:

-P1 15V power supply and current measurement interface

Connector P1: (Primary current measurement to SMPC regulation board P3)

Pin:	Name	I/O	Description & Specification
1	A+15V	O	+15V power supply; Used for interlock signaling.
2	A-15V	I	-15V power supply.
3	OUT	O	Output current representing the primary current.
4	A+15V	I	+15V power supply.

6.3.6 IGBT driver board

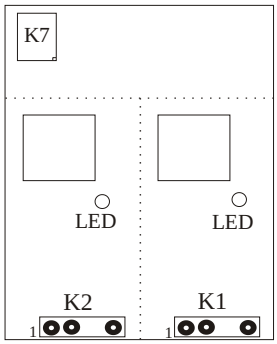


Fig. 6.4.6-1 IGBT driver board

The IGBT driver board has the following connectors:

-K1 Connectors for IGBT1
-K2 Connectors for IGBT2
-K7 Drive signals for one IGBT pair and status read backs

Connector K1: (Connector to IGBT1)

Pin:	Name	I/O	Description & Specification
1	E	O	Emitter IGBT1
2	G	O	Gate IGBT1.
3	C	O	Collector IGBT1.

Connector K2: (Connector to IGBT2)

Pin:	Name	I/O	Description & Specification
1	E	O	Emitter IGBT2
2	G	O	Gate IGBT2.
3	C	O	Collector IGBT2.

Connector K7: (From SMPC Regulation board connector P5 or P6)

Pin:	Name	I/O	Description & Specification
1	A+15	I	+15V Power supply.
2	AGND	I/O	Return 15V Power supply.
3	DR-fail A	O	Driver failure on driver A.
4	Gate A	I	Gate signal driver A
5	Gate B	O	Driver failure on driver B
6	DR-fail B	I	Gate signal driver B
7	B+5	I	+5V Power supply.

7 Spare parts

Please contact DANFYSIK A/S for a list of selected spare parts for your specific power supply.

8 Drawings

9 Change notes

10 Parts lists